GEOTECHNICAL EXPLORATION
COMBINED SEWER OVERFLOW SEPARATION
ATCHISON, KANSAS

Prepared for:

AFFINIS CORP
OVERLAND PARK, KANSAS

Prepared by:

GEOTECHNOLOGY, LLC DBA UES
OVERLAND PARK, KANSAS

Date:

MARCH 6, 2025

UES Project No.:

J045885.01

SAFETY
TEAMWORK
RESPONSIVENESS
INTEGRITY
VALUE
EXCELLENCE

Environmental
Geotechnical Engineering
Materials Testing
Field Inspections & Code Compliance
Geophysical Technology

March 6, 2025

Mr. Samuel Briscoe, P.E.
Affinis Corp
8900 Indian Creek Parkway, Suite 450
Corporate Woods Building 6
Overland Park, Kansas 66210

Re: Geotechnical Exploration
Combined Sewer Overflow Separation
Atchison, Kansas
UES Project No. J045885.01

Dear Mr. Briscoe:

Geotechnology, LLC dba UES (UES) presents in this report the results of the geotechnical exploration completed for the referenced project. This report includes our project understanding, observed site conditions, conclusions and/or recommendations, and support data as given in the Table of Contents.

We appreciate the opportunity to provide geotechnical services for this project. If you have questions regarding this report, or if we may be of any additional service to you, please contact the undersigned.

Sincerely,

UES



March 6, 2025

Matt McQuality, P.E.
Office Lead

MHMJAW:mhm

TABLE OF CONTENTS

Page

1.0 Introduction 1

2.0 Project Information 1

3.0 Geotechnical Exploration 1

4.0 Laboratory Testing 2

5.0 Subsurface Conditions 3
- 5.1 Pavement Section 3
- 5.2 Stratigraphy 3
- 5.3 Groundwater 3

6.0 Conclusions and Recommendations 4
- 6.1 Features Impacting Excavation 4
- 6.2 Site Grading 4
- 6.3 Manholes 5
- 6.4 Temporary Excavations 6
- 6.5 Temporary Shoring 6
- 6.6 Dewatering 7
- 6.7 Seismicity 7
- 6.8 Lateral Earth Pressures 8
- 6.9 Corrosivity 9

7.0 Recommended Additional Services 9

8.0 Limitations 10

FIGURES

Site Location and Topography 1
Aerial Photograph of Site and Boring Locations 2

APPENDICES

Important Information about This Geotechnical-Engineering Report ..... A
Logs of Borings B-1 through -3 ..... B
Boring Log: Terms and Symbols
Summary of Laboratory Results and Laboratory Test Computation Sheets C
Corrosion Potential Test Results ..... D
Pavement Core Photographs ..... E
Geotechnical Design Value Computation Sheets ..... F

GEOTECHNICAL EXPLORATION
COMBINED SEWER OVERLFOW SEPARATION
ATCHISON, KANSAS
MARCH 6, 2025 | UES PROJECT NO. J045885.01

1.0 INTRODUCTION

Geotechnology, LLC dba UES (UES) prepared this geotechnical exploration report for Affinis Corp (Affinis) for the design of storm sewer improvements in Atchison, Kansas. The services documented in this report were provided in general accordance with the scope of services described in UES's Proposal P045885.01, dated May 22, 2024. A representative of Affinis authorized our services by issuance of a Task Order last signed November 5, 2024.

The purposes of the geotechnical exploration were to evaluate the general subsurface profile and the engineering properties of the soil at the site, and to develop recommendations for the geotechnical aspects of the design and construction of the project as defined in our proposal. Our scope of services included three geotechnical borings, laboratory testing, engineering analyses, and preparation of draft and final reports.

2.0 PROJECT INFORMATION

The overall project includes the separation of combined sanitary and storm sewers in Atchison, Kansas. The project area includes (1) 15th Street between Atchison Street on the north and an unnamed alley on the south and (2) 16th Street between Main Street on the south and an unnamed alley on the north. Planned improvements include approximately 1,250 lineal feet of storm sewer and eight (8) stormwater structures. Other than the backfilling of excavations, site grading is not anticipated. Details of the project are being developed.

The storm sewer improvements parallel two-lane, asphalt-paved streets in a residential neighborhood. The site location and general topography of the area as per the 2018 U.S.G.S. map of the vicinity are shown on Figure 1.

3.0 GEOTECHNICAL EXPLORATION

The geotechnical exploration consisted of drilling three borings, designated as Borings B-1 through -3, at approximately the locations shown on Figure 2. The borings were located in areas accessible to the drill rig and free of utilities. A representative of UES estimated the boring locations using a handheld GPS device. The surface elevations at the boring locations, which are listed on the boring logs provided in Appendix B, were estimated using a USGS website. A licensed surveyor should measure the boring locations if accurate location data is required.

Each boring was drilled using a track-mounted CME 55 rotary drill rig equipped with an approximate 4-inch outside-diameter (O.D.) core barrel and 4-inch O.D. solid stem augers. Sampling of the overburden soils was accomplished at the depths indicated on the boring logs with 2-inch O.D. split-spoon or 3-inch O.D. thin-walled Shelby tube samplers, in general

accordance with the procedures outlined by ASTM D1586 and ASTM D1587, respectively. Standard Penetration Tests (SPTs) using a split-spoon sampler were performed to obtain the N-value[1] of the sampled material. Each borehole was backfilled with dry premix concrete and patched with quickset cement.

An engineer from UES provided direction during the field exploration, observed drilling and sampling, assisted in obtaining samples, and prepared field logs of the material encountered. A professional engineer has edited the final boring logs to incorporate results of the laboratory tests.

Unless noted on the boring logs, the lines designating the changes between various strata represent approximate boundaries. The transition between materials could be gradual or could occur between recovered samples. The stratification given on the boring logs, or described herein, is for use by UES in its analyses and should not be used as the basis of design or construction cost estimates without realizing that there can be variation from that shown or described.

The boring logs and related information depict subsurface conditions only at the specific locations and time where sampling was conducted. The passage of time could result in changes in conditions, interpreted to exist, at or between the locations where sampling was conducted.

4.0 LABORATORY TESTING

Laboratory testing was performed to estimate engineering and index properties of the sampled material. The following tests were performed (the corresponding ASTM standard, if applicable, is in parentheses):

- Moisture content (ASTM D 2216)
- Atterberg limits, single-point (ASTM D 4318)
- Unconfined compressive strength (ASTM D 2166)
- Corrosion potential tests (i.e., sulfide, pH, oxidation/reduction potential, resistivity, chloride and sulfate)

The results of the moisture content, Atterberg limits and unconfined compressive strength tests are presented on the boring logs; a tabular summary of these results and the corresponding laboratory computation sheets are presented in Appendix C. Corrosion potential test results are tabulated in Table 7 and included as Appendix D. Photographs of the pavement cores are presented in Appendix E.

[1] The Standard Penetration Test value, or N-value, is defined as the number of blows required to drive the split-spoon sampler 12 inches with a 140-pound hammer falling 30 inches. Since the split-spoon sampler is driven 18 inches or until refusal, the blows for the first 6 inches are for seating the sampler, and the number of blows for the final 12 inches is the N-value. Additionally, "refusal" of the split-spoon sampler occurs when the sampler is driven less than 6 inches with 50 blows of the hammer.

5.0 SUBSURFACE CONDITIONS

Presented herein is a summary of the pavement section, stratigraphy and groundwater encountered at the boring locations. The boring logs should be reviewed for further description of the surface and subsurface conditions at the boring locations.

5.1 Pavement Section

A summary of the pavement section at the boring locations is presented in Table 1.

Table 1. Pavement Section Summary

Boring	Asphalt, inches	Brick, inches	Concrete, inches	Crushed Rock, inches
B-1	2.5	None	10	None
B-2	3.5	2.5	2	None
B-3	1.5	None	5	None

5.2 Stratigraphy

Boring B-1 (15th Street and Atchison Street). Below the pavement section the stratigraphy consists of native, brown and gray, medium stiff to very stiff, lean clay some sand to a depth of approximately 12 feet underlain by dark gray and brown, very stiff, lean clay some sand, trace gravel to the 15-foot-depth of exploration. Auger refusal material was not encountered.

B-2 (15th Street and Kansas Avenue). Below the pavement section the stratigraphy consists of fill to a depth of approximately 7 feet. The fill sampled included dark brown and brown fat clay, gravel and concrete fragments. Below the fill, the soil stratigraphy consisted of native, stiff, brown lean clay to a depth of approximately 12 feet underlain by stiff, brown, fat clay to the 15-foot-depth of exploration. Auger refusal material was not encountered.

B-3 (16th Street and Main Street). Below the pavement section the stratigraphy consists of native, brown, stiff to very stiff, lean clay some sand to a depth of approximately 7 feet underlain by brown, medium dense, fine sand to a depth of approximately 9.5 feet underlain, in turn, by dark gray and brown, very stiff, lean clay some sand and gravel to a depth of approximately 12 feet. Below the clay, medium dense, brown, fine sand occurs to a depth of approximately 14.5 feet underlain by weathered shale to the 15-foot-depth of exploration. Auger refusal material was not encountered.

5.3 Groundwater

Groundwater levels observed at the borings during drilling are presented in Table 2. The presence or lack of groundwater levels might not represent present or future levels. Groundwater levels might vary substantially over time due to the effects of seasonal variation in precipitation, recharge, levels of nearby tributaries, or other factors not evident at the time of exploration. Therefore, groundwater levels during construction or at other times in the life of the structures might differ than the levels indicated on the boring logs. Water might be trapped within permeable fill pockets, if present, and in utility trenches backfilled with clean rock.

Table 2. Groundwater Summary

Boring	Surface Elevation	Approx. Groundwater Level During Drilling	
		Feet	Elev.
B-1	El 923[2]	10	El 913
B-2	El 872	Not Observed	
B-3	El 860	13.5	El 846.5

6.0 CONCLUSIONS AND RECOMMENDATIONS

UES should be allowed to review final plans to check that our recommendations have been correctly implemented and are applicable to final design. A discussion of the geotechnical features that might influence construction are presented in the following paragraphs.

6.1 Features Impacting Excavation

Existing Fill. Fill was encountered in one boring. Fill and trench backfill likely occurs in unexplored areas of the site. Fill might extend to greater depths than those depicted on the boring logs. The composition of the fill could vary in unexplored locations. In the absence of documentation, fill and trench backfill should be considered uncontrolled and unstable. Unstable fill encountered in excavations could collapse. As discussed herein, shoring might be required for temporary excavations.

Groundwater. Groundwater was encountered in two of the borings. Dewatering should be provided to maintain groundwater at least 12 inches below the bottom of excavations to maintain a stable base for construction and to reduce instability of excavated slopes. Further discussion on dewatering is presented herein.

Sand. Sand strata was encountered in one boring. Sand might occur in unexplored areas of the site. Sand is collapsible. As discussed herein, shoring might be required for excavations in sand.

6.2 Site Grading

Suitable Fill Materials. On-site materials generated from excavations are expected to include fine-grained soil, sand and pieces of asphalt, concrete and brick. Fine-grained soil and sand are typically suitable for trench backfill provided the material is moisture conditioned and free of deleterious materials. Pieces of asphalt, concrete and brick should be processed to form a well-graded mixture with a maximum 2-inch particle size. Alternatively, pieces of asphalt, concrete and brick could be hauled of the site and properly disposed.

Imported backfill should consist of lean, fine-grained soil (liquid limit less than 45) or well-graded crushed limestone with a 2-inch maximum particle size. Acceptable fill soils include non-organic materials designated as CL, CL-ML, ML, GM, GW and GW-GM in accordance with ASTM D 2487. Frozen soils and soil with deleterious materials should not be used for trench backfill.

[2] Elevations herein are in feet and reference 1988 North American Vertical Datum (NAVD88)

Pipe Bedding. Soft soil and disturbed soil subgrade areas should be replaced with compacted, well-graded crushed rock. Well-graded crushed rock should be used as the bedding material for pipes and sewer manholes. Loose material that accumulates in excavations should be removed prior to laying pipe or placing concrete or rock base for manholes.

Trench Backfill Placement. Trench backfill should be placed in uniform lifts and compacted. The loose lift thickness should not exceed 6 inches. The fill should be systematically compacted to the levels given in Table 3. The soil should be placed at moisture contents compatible with the required unit weight. Depending on the soil moisture at the time of construction, aeration or wetting might be required to achieve compaction. Deleterious material should not be included in backfill, and backfill should not be placed on soft materials or frozen ground.

Table 3. Compaction Summary

Category	Minimum Compaction*
Fine-Grained Soil	90%
Well-Graded Crushed Limestone	95%

* Measured as a percent of the maximum unit weight as determined by the modified Proctor test (ASTM D 1557).

Trench Backfill. Settlement of trench backfill can result in unsightly depressions and localized failures. The magnitude of settlement can be substantially reduced by mechanical compaction of the trench backfill. Clean rock should not be used for trench backfill. Clean rock can collect water which can soften the underlying fine-grained soils or result in the migration of fines and loss of subgrade support.

Subgrade Protection. Drainage of the construction areas should be provided to protect excavations and structure subgrades from the detrimental effects of weather conditions during construction. Finished subgrades and structure excavations should be kept free of standing water. Concrete should be placed in foundation excavations the same day they are completed.

Collection and Disposal of Site Water. Control of surface runoff should be maintained in compliance with the rules and regulations set forth in the Federal Water Pollution Control Act. Additionally, permits related to site grading activities and control of storm water during construction activities should be obtained from the applicable governmental jurisdiction(s).

6.3 Manholes

Net Allowable Bearing Pressure. Manholes bearing on the natural soil and compacted fill may be designed for a net allowable bearing pressure of 1,500 pounds per square foot (psf). Areas or subgrades which exhibit soft or unstable characteristics should be overexcavated and backfilled with compacted well-graded crushed rock, lean concrete or concrete.

Settlement of new manholes is expected to be less than 1/4 inch provided the manholes are proportioned and constructed as recommended herein. Estimated values of settlement contained in this report are based on our experience with projects of a similar nature. Consolidation tests and corresponding settlement calculations have not been made.

Uplift Resistance. Uplift loads can be resisted by the buoyant weight of the manhole and frictional resistance between the manhole exterior and compacted backfill placed against the manhole. For compacted backfill placed against a manhole, an allowable frictional resistance value of 500 psf may be used. The frictional resistance in the top 36 inches of compacted fill should be neglected due to seasonal variations in moisture and frost penetration. The friction value provided in this section does not need to be reduced for the portion of a structure below groundwater.

6.4 Temporary Excavations

If site geometry permits, it might be possible to lay slopes back to a stable configuration. Temporary excavation slopes should be consistent with safety regulations. Worker safety and classification of soil type is the responsibility of the contractor. The materials encountered during excavations for the proposed project are anticipated to consist of fill of unknown origin; medium stiff to very stiff clay, and sand. Native medium stiff to very stiff clay soil can generally be classified as OSHA Type B soil. Fill of unknown origin and sand should be considered OSHA Type C soil. Temporary slopes per OSHA regulations for these soil types are presented in Table 4.

Table 4. Temporary Slopes Per OSHA Regulations

Soil Type	Slope Inclination[a]
B	1V:1H or flatter
C[b]	1V:1.5H or flatter; benching not permitted
Soil Slopes Below Groundwater (Regardless of soil type)	

[a] Sloping or benching for excavations greater than 20 feet deep shall be designed by a registered professional engineer.

[b] Includes excavations terminating in Type C soil.

The contractor should be aware that excavation depths and inclinations (including adjacent existing slopes) should not exceed those specified in local, state or federal safety regulations, e.g., OSHA Health and Safety Standards for Excavations, 29 CFR, or successor regulations. Such regulations are strictly enforced and, if not followed, the contractor, or earthwork or utility subcontractors could be subjected to substantial penalties. Construction site safety is the sole responsibility of the contractor, who shall also be solely responsible for the means, methods and sequencing of construction operations.

Temporary slopes left open might undergo sloughing and result in an unstable situation. The contractor should evaluate stability and failure consequences before open cut slopes are made. Minor sloughing of open face slopes might occur. If the slope is expected to remain open for an extended time, an impermeable membrane covering the slopes could be considered as a means to reduce the potential for slope degradation and instability. It is important to note that soils encountered in the construction excavations might vary across the sites, and that even if the OSHA criteria are used, there is a potential for slope failure.

6.5 Temporary Shoring

Excavated temporary slopes might not be feasible at all locations and temporary shoring systems might be required. While different temporary shoring types and configurations can be used, the

more common types that are applicable to this site are trench boxes, and soldier pile and lagging. Brief discussions of potential temporary shoring methods are presented in the following paragraphs. Other methods of temporary shoring, however, may also be considered. Temporary shoring installed adjacent to structures sensitive to movement should be designed for at-rest lateral earth pressures. Temporary shoring permitted to tilt out approximately 0.5 percent of the depth of the excavation may be designed for active lateral earth pressures.

Trench Boxes. Trench boxes are steel or aluminum structures used for protecting personnel performing duties within a trench. Trench boxes are customarily constructed with sidewalls of varying thicknesses, are held apart by internal steel or aluminum bracing, and are placed within a trench after excavation. Trench boxes are typically limited to excavation depths of 10 to 12 feet. Since voids typically occur between the sidewalls of the trench box and the edge of the excavation, trench boxes do not prevent collapse of excavation sidewalls. In addition, trench boxes are not recommended for excavations adjacent to structures sensitive to movement.

Soldier Piles and Lagging. Construction of soldier piles involves placing steel H-piles in predrilled holes that are subsequently backfilled with flowable fill or lean concrete. As the excavation proceeds downward, timber lagging or steel sheeting is installed between the exposed H-piles to restrain the exposed cohesive soil. Additional lateral resistance can be obtained by installing internal bracing or tiebacks. Voids between the timber lagging or steel sheeting should be filled.

6.6 Dewatering

Depending on the level of groundwater at the time of construction, it might be necessary to dewater excavations to enable construction to be carried out under relatively dry conditions, to prevent subgrade disturbance, and to provide stability of excavated slopes. Groundwater should be lowered to at least 12 inches below excavation bottoms. It is anticipated that groundwater in excavations in fine-grained soil can be controlled by use of sumps and pumps. Control of groundwater in excavations that extend into the underlying sand might require single or multistage well points using centrifugal, vacuum or jet-eductor pumps. Selection of a dewatering approach depends on the season of the year, volume of water anticipated, type of subgrade soils, proximity of source and other factors. If sumps cannot control the groundwater seepage, UES should be contacted to review the method of dewatering proposed by the contractor. Dewatering systems should be designed by a contractor with experience in similar soil conditions.

6.7 Seismic Design Coefficients

Per the general procedures of Section 1613 of IBC 2021, the soil profile at the project site can be defined as Class D – Stiff Soil. Based on the web application *Seismic Design Maps* prepared by the Structural Engineers Association of California (SEAOC) and the California's Office of Statewide Health Planning and Development (OSHPD) using the latitude and longitude coordinates of the project site, the seismic parameters that can be used for this project are given in Table 5.

Table 5. Seismic Parameters

Seismic Site Class	S_s (g)	S_1 (g)	S_{MS} (g)	S_{M1} (g)	S_{DS} (g)	S_{D1} (g)
D – Stiff Soil	0.089	0.061	0.143	0.147	0.095	0.098

6.8 Lateral Earth Pressures

Below-grade walls should be designed to withstand lateral earth pressures caused by adjacent soil and surcharge load. Design lateral pressures from surcharge loads should be added to the lateral earth pressure load. Lateral earth pressures can vary with wall restraint conditions, type of backfill and method of backfill compaction.

Design values are provided in Table 6 for lateral loads on below-grade walls. If tilting or deflection is not tolerable, at-rest earth pressures should be assumed. Computation sheets for the geotechnical design values are presented as Appendix F.

Table 6. Lateral Earth Pressures for Horizontal Ground Surfaces – Drained Condition

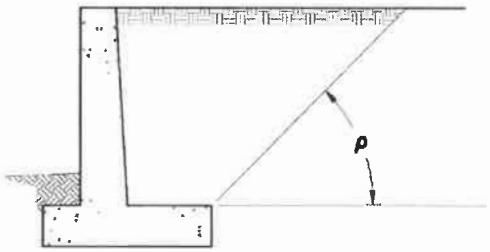
Description of Backfill	Design Soil Lateral Load (psf per foot of depth)	
	At-rest	Active
Fine-Grained Soil	65h + 0.55q	45h + 0.38q
Well-Graded Crushed Stone (GW)	56h + 0.40q	35h + 0.25q

Where:

h = depth below adjacent grade, feet
q = surcharge load, psf

The given values assume that hydrostatic pressures will not develop behind the walls and that the wall backfill will be compacted as previously recommended. Therefore, below-grade walls should have perforations to allow for dissipation of water pressure against the walls. Undrained walls might be subjected to additional pressures from groundwater, perched water, or surface water infiltration.

For the equations in Table 6 to be valid for crushed stone backfill, the backfill should be placed in a wedge extending upward and away from the edge of the footings at an angle 'ρ' of 45 degrees or flatter (see the following figure). If crushed stone is to be placed within a wedge at an angle 'ρ' of more than 45 degrees from the bottom of the structure, the values for fine-grained soil provided in Table 6 should be used.



6.9 Corrosivity

Iron Pipe. The Ductile Iron Pipe Association (DIPRA) 10-point corrosion potential evaluation system was used to assess the corrosion potential of a soil sampled collected at Boring B-2. The evaluation procedure requires the following analytical tests: soil resistivity, pH, oxidation-reduction potential, sulfides and moisture. For a given soil sample, each parameter is evaluated and assigned points for all five areas. The points are totaled and if the sum is 10 or more, the soil is considered potentially corrosive to iron pipe, which would warrant protective measures. Corrosion potential test results are summarized in Table 7.

Table 7. Summary of Corrosion Potential Test Results

Sample	Resistivity, ohms-cm	pH	Redox Potential, mv	Sulfide, mg/kg	Moisture, percent	Chloride, mg/kg	Sulfate, mg/kg
B-2, 10 feet	225	5.8	231.9	ND	19	507	476

ND – Not detected

The DIPRA point total for the tested sample is presented in Table 8; the tested sample had a point total of more than 10. Consequently, protective measures such as cathodic protection or a proprietary coating for buried metal features are warranted based on the results of the analytical tests.

Table 8. DIPRA Point Total

Boring	Location	Sample Description	Depth, Feet	DIPRA Point Total
B-2	15th Street south of Kansas Avenue	Stiff, brown, lean CLAY – (CL)	10	11

Cement. Based on the results of the analytical tests, below-grade concrete will be exposed to a moderate sulfate condition and Type II cement should be used.

7.0 RECOMMENDED ADDITIONAL SERVICES

The conclusions and recommendations given in this report are based on: UES's understanding of the proposed design and construction, as outlined in this report; site observations; interpretation of the exploration data; and our experience. Since the intent of the design recommendations is best understood by UES, we recommend that UES be included in the final design and construction process and be retained to review the project plans and specifications to confirm that the

recommendations given in this report have been correctly implemented. We recommend that UES be retained to participate in prebid and preconstruction conferences to reduce the risk of misinterpretation of the conclusions and recommendations in this report relative to the proposed construction of the subject project.

Since actual subsurface conditions between boring locations may vary from those encountered in the borings, our design recommendations are subject to adjustment in the field based on the subsurface conditions encountered during construction. Therefore, we recommend that UES be retained to provide construction observation services as a continuation of the design process to confirm the recommendations in this report and to revise them accordingly to accommodate differing subsurface conditions. Construction observation is intended to enhance compliance with project plans and specifications. It is not insurance, nor does it constitute a warranty or guarantee of any type. Regardless of construction observation, contractors, suppliers, and others are solely responsible for the quality of their work and for adhering to plans and specifications.

8.0 LIMITATIONS

This report has been prepared on behalf of, and for the exclusive use of, the client for specific application to the named project as described herein. If this report is provided to other parties, it should be provided in its entirety with all supplementary information. In addition, the client should make it clear that the information is provided for factual data only, and not as a warranty of subsurface conditions presented in this report.

UES has attempted to conduct the services reported herein in a manner consistent with that level of care and skill ordinarily exercised by members of the profession currently practicing in the same locality and under similar conditions. The recommendations and conclusions contained in this report are professional opinions. The report is not a bidding document and should not be used for that purpose.

Our scope of service for this phase of the project did not include any environmental assessment or investigation for the presence or absence of wetlands or hazardous or toxic materials in the soil, surface water, groundwater, or air, on or below or around this site. Any statements in this report or on the boring logs regarding odors noted or unusual or suspicious items or conditions observed are strictly for the information of our client. Our scope of service did not include an assessment of the effects of flooding and erosion of creeks or rivers adjacent to or on the project site.

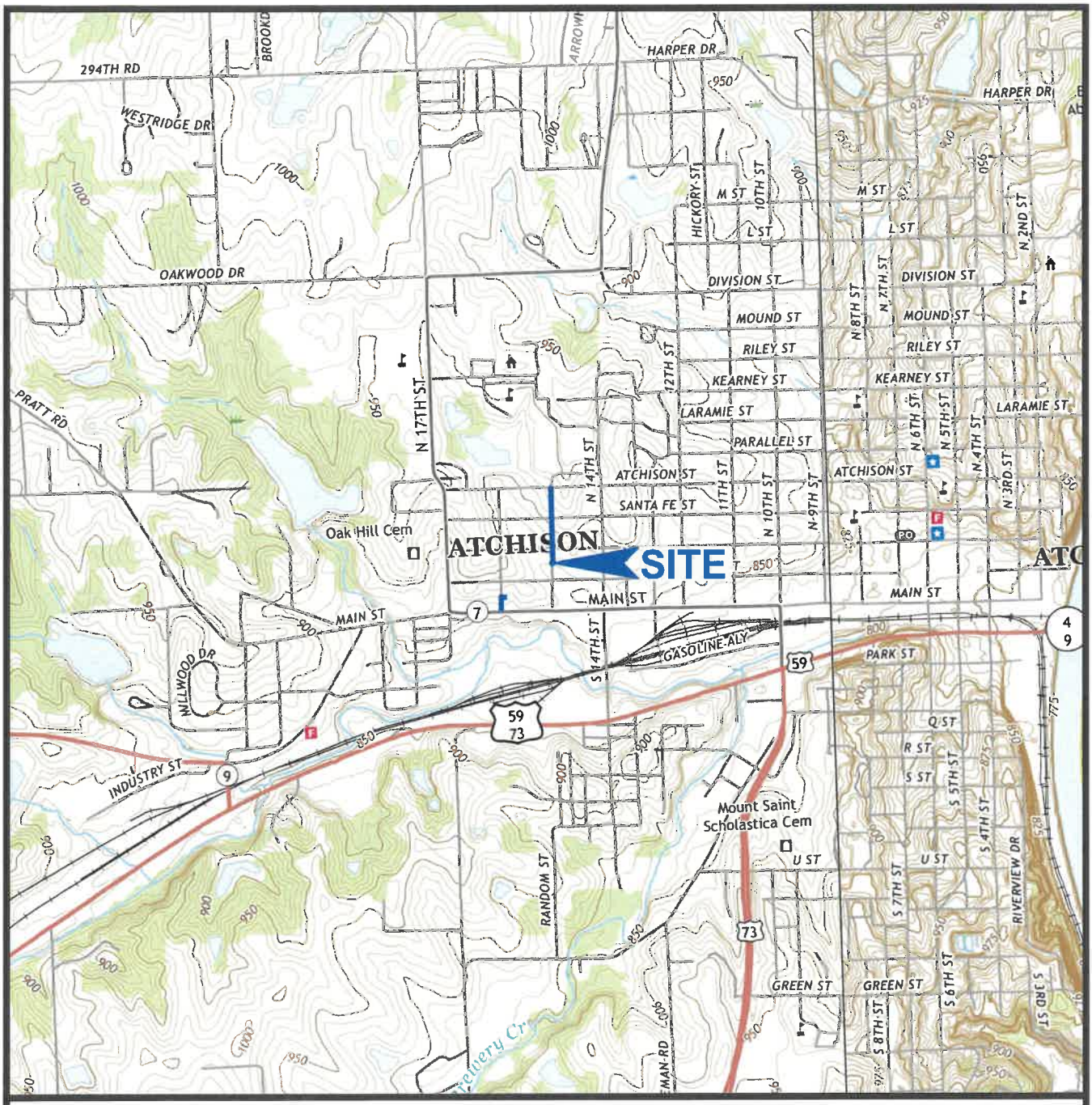
The analyses, conclusions, and recommendations contained in this report are based on the data obtained from the subsurface exploration. The field exploration methods used indicate subsurface conditions only at the specific locations where samples were obtained, only at the time they were obtained, and only to the depths penetrated. Consequently, subsurface conditions may vary gradually, abruptly, and/or nonlinearly between sample locations and/or intervals.

The conclusions or recommendations presented in this report should not be used without UES's review and assessment if the nature, design, or location of the facilities is changed, if there is a

substantial lapse in time between the submittal of this report and the start of work at the site, or if there is a substantial interruption or delay during work at the site. If changes are contemplated or delays occur, UES must be allowed to review them to assess their impact on the findings, conclusions, and/or design recommendations given in this report. UES will not be responsible for any claims, damages, or liability associated with any other party's interpretations of the subsurface data or with reuse of the subsurface data or engineering analyses in this report.

The recommendations included in this report have been based in part on assumptions about variations in site stratigraphy that may be evaluated further during sewer construction. UES should be retained to perform construction observation and continue its geotechnical engineering service using observational methods. UES cannot assume liability for the adequacy of its recommendations when they are used in the field without UES being retained to observe construction.

A copy of "Important Information about This Geotechnical-Engineering Report" that is published by the Geotechnical Business Council (GBC) of the Geoprofessional Business Association (GBA) is included in Appendix A for your review. The publication discusses some other limitations, as well as ways to manage risk associated with subsurface conditions.



NOTES

1. Plan adapted from 7.5 minute U.S.G.S. maps for Atchison West and Atchison East, Kansas quadrangles, last revised in 2018.

0 2,000 4,000

SCALE IN FEET

N

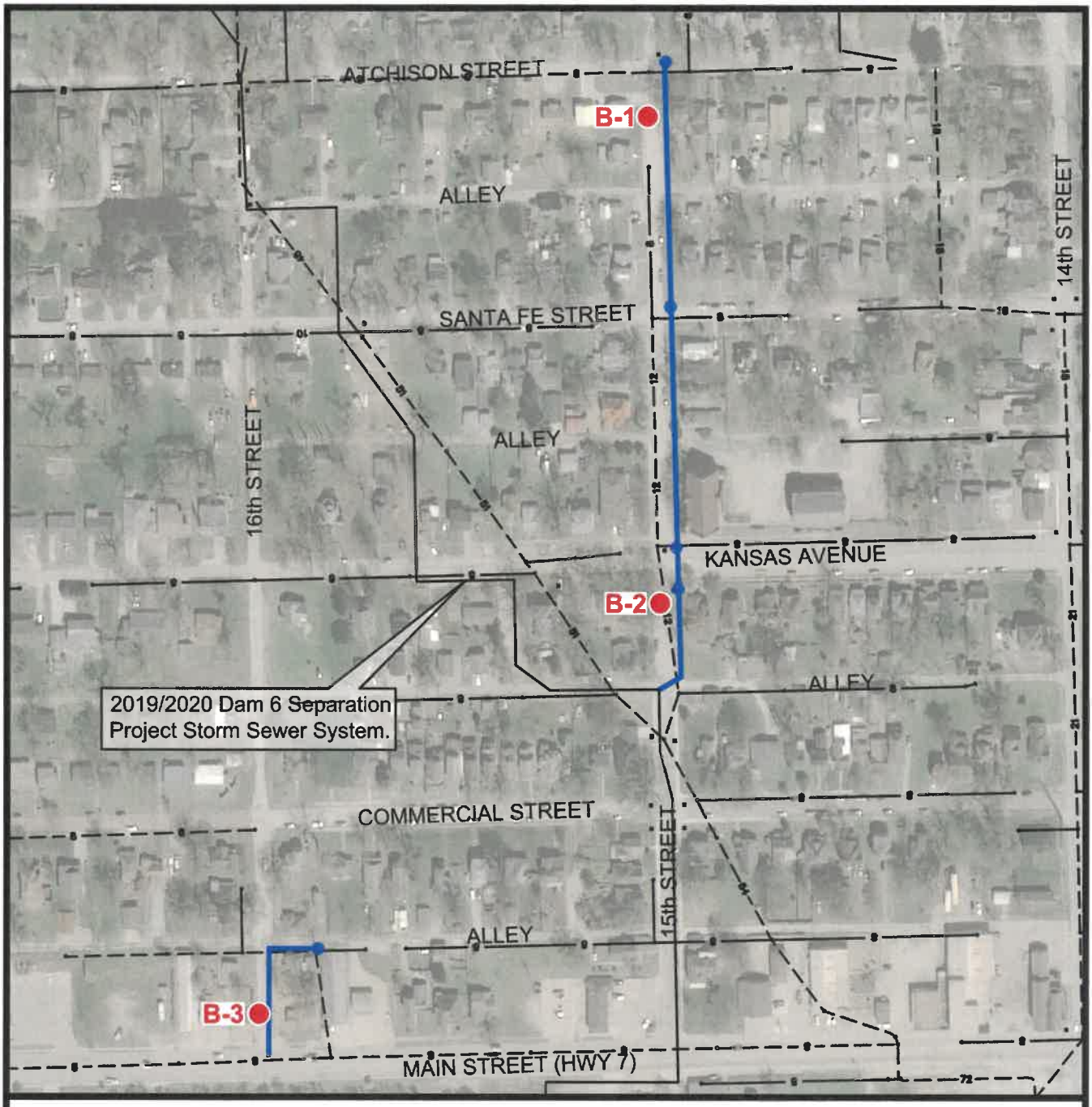
Drawn By: WAH	Ck'd By: MHM	App'vd By: JAW
Date: 12-13-24	Date: 1-16-25	Date: 1-21-25

UES™

Combined Sewer Overflow Separation
Atchison, Kansas

SITE LOCATION AND TOPOGRAPHY

Project Number J045885.01	FIGURE 1



NOTES

1. Plan adapted from an April 6, 2020 aerial photograph courtesy of Google Earth and an undated drawing titled "Exhibit 3 - 14th Street Sewershed Separation (Phase 3)", supplied by the client.
2. Borings were located in the field utilizing a hand-held GPS device.

LEGEND

- Boring Location
- Proposed Project Alignment
- Storm Sewer Inlet
- Existing Separate Sewer
- Existing Combined Sewer

N

0 250 500

SCALE IN FEET

Drawn By: WAH	Ck'd By: MHM	App'vd By: JAW
Date: 12-13-24	Date: 1-16-25	Date: 1-21-25

UES

Combined Sewer Overflow Separation
Atchison, Kansas

AERIAL PHOTOGRAPH OF SITE AND BORING LOCATIONS

Project Number J045885.01	FIGURE 2

APPENDIX A

Important Information about This Geotechnical-Engineering Report

Important Information about This

Geotechnical-Engineering Report

Subsurface problems are a principal cause of construction delays, cost overruns, claims, and disputes.

While you cannot eliminate all such risks, you can manage them. The following information is provided to help.

Geotechnical Services Are Performed for Specific Purposes, Persons, and Projects

Geotechnical engineers structure their services to meet the specific needs of their clients. A geotechnical-engineering study conducted for a civil engineer may not fulfill the needs of a constructor — a construction contractor — or even another civil engineer. Because each geotechnical- engineering study is unique, each geotechnical-engineering report is unique, prepared *solely* for the client. No one except you should rely on this geotechnical-engineering report without first conferring with the geotechnical engineer who prepared it. *And no one — not even you* — should apply this report for any purpose or project except the one originally contemplated.

Read the Full Report

Serious problems have occurred because those relying on a geotechnical-engineering report did not read it all. Do not rely on an executive summary. Do not read selected elements only.

Geotechnical Engineers Base Each Report on a Unique Set of Project-Specific Factors

Geotechnical engineers consider many unique, project-specific factors when establishing the scope of a study. Typical factors include: the client's goals, objectives, and risk-management preferences; the general nature of the structure involved, its size, and configuration; the location of the structure on the site; and other planned or existing site improvements, such as access roads, parking lots, and underground utilities. Unless the geotechnical engineer who conducted the study specifically indicates otherwise, do not rely on a geotechnical-engineering report that was:

- not prepared for you;
- not prepared for your project;
- not prepared for the specific site explored; or
- completed before important project changes were made.

Typical changes that can erode the reliability of an existing geotechnical-engineering report include those that affect:

- the function of the proposed structure, as when it's changed from a parking garage to an office building, or from a light-industrial plant to a refrigerated warehouse;
- the elevation, configuration, location, orientation, or weight of the proposed structure;
- the composition of the design team; or
- project ownership.

As a general rule, *always* inform your geotechnical engineer of project changes—even minor ones—and request an assessment of their impact. *Geotechnical engineers cannot accept responsibility or liability for problems that occur because their reports do not consider developments of which they were not informed.*

Subsurface Conditions Can Change

A geotechnical-engineering report is based on conditions that existed at the time the geotechnical engineer performed the study. *Do not rely on a geotechnical-engineering report whose adequacy may have been affected by:* the passage of time; man-made events, such as construction on or adjacent to the site; or natural events, such as floods, droughts, earthquakes, or groundwater fluctuations. *Contact the geotechnical engineer before applying this report to determine if it is still reliable.* A minor amount of additional testing or analysis could prevent major problems.

Most Geotechnical Findings Are Professional Opinions

Site exploration identifies subsurface conditions only at those points where subsurface tests are conducted or samples are taken. Geotechnical engineers review field and laboratory data and then apply their professional judgment to render an opinion about subsurface conditions throughout the site. Actual subsurface conditions may differ — sometimes significantly — from those indicated in your report. Retaining the geotechnical engineer who developed your report to provide geotechnical-construction observation is the most effective method of managing the risks associated with unanticipated conditions.

A Report's Recommendations Are Not Final

Do not overrely on the confirmation-dependent recommendations included in your report. *Confirmation-dependent recommendations are not final,* because geotechnical engineers develop them principally from judgment and opinion. Geotechnical engineers can finalize their recommendations *only* by observing actual subsurface conditions revealed during construction. *The geotechnical engineer who developed your report cannot assume responsibility or liability for the report's confirmation-dependent recommendations if that engineer does not perform the geotechnical-construction observation required to confirm the recommendations' applicability.*

A Geotechnical-Engineering Report Is Subject to Misinterpretation

Other design-team members' misinterpretation of geotechnical-engineering reports has resulted in costly

problems. Confront that risk by having your geotechnical engineer confer with appropriate members of the design team after submitting the report. Also retain your geotechnical engineer to review pertinent elements of the design team's plans and specifications. Constructors can also misinterpret a geotechnical-engineering report. Confront that risk by having your geotechnical engineer participate in prebid and preconstruction conferences, and by providing geotechnical construction observation.

Do Not Redraw the Engineer's Logs

Geotechnical engineers prepare final boring and testing logs based upon their interpretation of field logs and laboratory data. To prevent errors or omissions, the logs included in a geotechnical-engineering report should *never* be redrawn for inclusion in architectural or other design drawings. Only photographic or electronic reproduction is acceptable, *but recognize that separating logs from the report can elevate risk.*

Give Constructors a Complete Report and Guidance

Some owners and design professionals mistakenly believe they can make constructors liable for unanticipated subsurface conditions by limiting what they provide for bid preparation. To help prevent costly problems, give constructors the complete geotechnical-engineering report, *but* preface it with a clearly written letter of transmittal. In that letter, advise constructors that the report was not prepared for purposes of bid development and that the report's accuracy is limited; encourage them to confer with the geotechnical engineer who prepared the report (a modest fee may be required) and/or to conduct additional study to obtain the specific types of information they need or prefer. A prebid conference can also be valuable. *Be sure constructors have sufficient time* to perform additional study. Only then might you be in a position to give constructors the best information available to you, while requiring them to at least share some of the financial responsibilities stemming from unanticipated conditions.

Read Responsibility Provisions Closely

Some clients, design professionals, and constructors fail to recognize that geotechnical engineering is far less exact than other engineering disciplines. This lack of understanding has created unrealistic expectations that have led to disappointments, claims, and disputes. To help reduce the risk of such outcomes, geotechnical engineers commonly include a variety of explanatory provisions in their reports. Sometimes labeled "limitations," many of these provisions indicate where geotechnical engineers' responsibilities begin and end, to help others recognize their own responsibilities and risks. *Read these provisions closely.* Ask questions. Your geotechnical engineer should respond fully and frankly.

Environmental Concerns Are Not Covered

The equipment, techniques, and personnel used to perform an *environmental* study differ significantly from those used to perform a *geotechnical* study. For that reason, a geotechnical-engineering report does not usually relate any environmental findings, conclusions, or recommendations; e.g., about the likelihood of encountering underground storage tanks or regulated contaminants. *Unanticipated environmental problems have led to numerous project failures.* If you have not yet obtained your own environmental information, ask your geotechnical consultant for risk-management guidance. *Do not rely on an environmental report prepared for someone else.*

Obtain Professional Assistance To Deal with Mold

Diverse strategies can be applied during building design, construction, operation, and maintenance to prevent significant amounts of mold from growing on indoor surfaces. To be effective, all such strategies should be devised for the *express purpose* of mold prevention, integrated into a comprehensive plan, and executed with diligent oversight by a professional mold-prevention consultant. Because just a small amount of water or moisture can lead to the development of severe mold infestations, many mold- prevention strategies focus on keeping building surfaces dry. While groundwater, water infiltration, and similar issues may have been addressed as part of the geotechnical- engineering study whose findings are conveyed in this report, the geotechnical engineer in charge of this project is not a mold prevention consultant; *none of the services performed in connection with the geotechnical engineer's study were designed or conducted for the purpose of mold prevention. Proper implementation of the recommendations conveyed in this report will not of itself be sufficient to prevent mold from growing in or on the structure involved.*

Rely, on Your GBC-Member Geotechnical Engineer for Additional Assistance

Membership in the Geotechnical Business Council of the Geoprofessional Business Association exposes geotechnical engineers to a wide array of risk-confrontation techniques that can be of genuine benefit for everyone involved with a construction project. Confer with you GBC-Member geotechnical engineer for more information.

GEOTECHNICAL BUSINESS COUNCIL
of the Geoprofessional Business Association

8811 Colesville Road/Suite G106, Silver Spring, MD 20910
Telephone: 301/565-2733 Facsimile: 301/589-2017
e-mail: info@geoprofessional.org www.geoprofessional.org

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APPENDIX B

Logs of Borings B-1 through -3
Boring Log: Terms and Symbols

Surface Elevation 923 | Completion Date: 12/5/24

Datum NAVD88

DEPTH IN FEET	DESCRIPTION OF MATERIAL	GRAPHIC LOG	DRY UNIT WEIGHT (pcf) SPT BLOW COUNTS CORE RECOVERY/RQD	SAMPLES
	ASPHALT - 2.5 inches			
	CONCRETE - 10 inches			
	CLAY - brown and gray, medium stiff to very stiff, lean, some sand - (CL)		3-3-4	SS1
			103	ST2
5			3-4-4	SS3
			4-5-6	SS4
10				
	CLAY - dark gray and brown, very stiff, lean, some sand, trace gravel - CL		5-8-12	SS5
15	Boring terminated at 15 feet.			
20				

SHEAR STRENGTH, tsf

Δ - UU/2 ○ - QU/2 □ - PP/2

0.5 1.0 1.5 2.0 2.5

STANDARD PENETRATION RESISTANCE

(ASTM D 1586)

▲ N-VALUE (BLOWS PER FOOT)

WATER CONTENT, %

PL 10 20 30 40 50 LL

NOTE: STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARIES BETWEEN SOIL TYPES AND THE TRANSITION MAY BE GRADUAL. GRAPHIC LOG FOR ILLUSTRATION PURPOSES ONLY

GROUNDWATER DATA

ENCOUNTERED AT 10 FEET ▽

DRILLING DATA

4" AUGER ___ HOLLOW STEM

WASHBORING FROM ___ FEET

DJT DRILLER RAW LOGGER

CME 55TRK DRILL RIG

HAMMER TYPE Auto

REMARKS:

Drawn by: RAW	Check by: MHM	App'vd by: JAW
Date: 12/6/24	Date: 1/16/25	Date: 1-21-25

UES

Combined Sewer Overflow Separation Atchison, Kansas

LOG OF BORING: B-1

Project No. J045885.01

Surface Elevation 872 | Completion Date: 12/5/24

Datum NAVD88

DEPTH IN FEET	DESCRIPTION OF MATERIAL	GRAPHIC LOG	DRY UNIT WEIGHT (pcf) SPT BLOW COUNTS CORE RECOVERY/RQD	SAMPLES
	ASPHALT - 3.5 inches			
	BRICK - 2.5 inches			
	CONCRETE - 2 inches			
	FILL - dark brown and brown fat clay, some gravel and concrete fragments poor sample recovery		4-3-5	SS1
			2-1-2	SS2
5				
			7-3-3	SS3
	CLAY - brown, stiff, lean - (CL)			
			103	ST4
10				
	CLAY - brown, stiff, fat - CH			
			3-4-5	SS5
15	Boring terminated at 15 feet.			
20				

SHEAR STRENGTH, tsf

Δ - UU/2 ○ - QU/2 □ - PP/2

0.5 1.0 1.5 2.0 2.5

STANDARD PENETRATION RESISTANCE

(ASTM D 1586)

▲ N-VALUE (BLOWS PER FOOT)

WATER CONTENT, %

PL 10 20 30 40 50 LL

NOTE: STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARIES BETWEEN SOIL TYPES AND THE TRANSITION MAY BE GRADUAL. GRAPHIC LOG FOR ILLUSTRATION PURPOSES ONLY.

GROUNDWATER DATA

X FREE WATER NOT ENCOUNTERED DURING DRILLING

DRILLING DATA

4" AUGER ___ HOLLOW STEM

WASHBORING FROM ___ FEET

DJT DRILLER RAW LOGGER

CME 55TRK DRILL RIG

HAMMER TYPE Auto

REMARKS:

Drawn by: RAW	Check by: MHM	App'vd by: JAW
Date: 12/6/24	Date: 1/16/25	Date: 1-21-25

UES™

Combined Sewer Overflow Separation Atchison, Kansas

LOG OF BORING: B-2

Project No. J045885.01

Surface Elevation 860 | Completion Date: 12/5/24

Datum NAVD88

DEPTH IN FEET	DESCRIPTION OF MATERIAL	GRAPHIC LOG	DRY UNIT WEIGHT (pcf) SPT BLOW COUNTS CORE RECOVERY/RQD	SAMPLES
	ASPHALT - 1.5 inches			
	CONCRETE - 5 inches (poor recovery)			
	CLAY - brown, stiff to very stiff, lean, some sand - (CL)		3-4-6	SS1
			5-9-11	SS2
5				
	SAND - brown, medium dense, fine - SP		7-9-9	SS3
			8-9-10	SS4
10	CLAY - dark gray and brown, very stiff, lean, some sand and gravel - CL			
	SAND - brown, medium dense, fine - SP ▽		3-5-8	SS5
	weathered shale			
15	Boring terminated at 15 feet.			
20				

SHEAR STRENGTH, tsf

Δ - UU/2 ○ - QU/2 □ - PP/2

0.5 1.0 1.5 2.0 2.5

STANDARD PENETRATION RESISTANCE

(ASTM D 1586)

▲ N-VALUE (BLOWS PER FOOT)

WATER CONTENT, %

PL 10 20 30 40 50 LL

NOTE: STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARIES BETWEEN SOIL TYPES AND THE TRANSITION MAY BE GRADUAL. GRAPHIC LOG FOR ILLUSTRATION PURPOSES ONLY.

GROUNDWATER DATA

ENCOUNTERED AT 13.5 FEET ▽

DRILLING DATA

4" AUGER ___ HOLLOW STEM

WASHBORING FROM ___ FEET

DJT DRILLER RAW LOGGER

CME 55TRK DRILL RIG

HAMMER TYPE Auto

REMARKS:

Drawn by: RAW	Check by: MHM	App'vd by: JAW
Date: 12/6/24	Date: 1/16/25	Date: 1-21-25

UES™

Combined Sewer Overflow Separation Atchison, Kansas

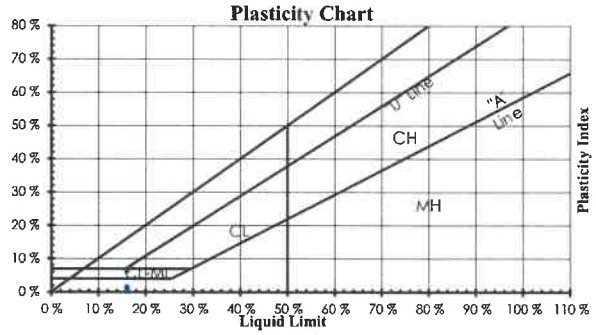
LOG OF BORING: B-3

Project No. J045885.01

BORING LOG: TERMS AND SYMBOLS

LEGEND

CS	Continuous Sampler
GB	Grab Sample
NQ	NQ Rock Core
PST	Three-Inch Diameter Piston Tube Sample
SS	Split-Spoon Sample (Standard Penetration Test)
ST	Three-Inch Diameter Shelby Tube Sample
*	Sample Not Recovered
PL	Plastic Limit (ASTM D4318)
LL	Liquid Limit (ASTM D4318)
SV	Shear Strength from Field Vane (ASTM D2573)
UU	Shear Strength from Unconsolidated-Undrained Triaxial Compression Test (ASTM D2850)
QU	Shear Strength from Unconfined Compression Test (ASTM D2166)



SOIL GRAIN SIZE

US STANDARD SIEVE

12"	3"	3/4"	4	10	40	200

BOULDERS	COBBLES	GRAVEL		SAND			SILT	CLAY
		COARSE	FINE	COARSE	MEDIUM	FINE		

300	76.2	19.1	4.76	2.00	0.42	0.074	0.005

SOIL GRAIN SIZE IN MILLIMETERS

UNIFIED SOIL CLASSIFICATION SYSTEM

Major Divisions			*Symbol*	*Description*
Coarse-Grained Soils (More than 50% Larger than No. 200 Sieve Size)	Gravel and Gravelly Soil	Clean Gravels Little or no Fines	GW	Well-Graded Gravel, Gravel- Sand Mixture
			GP	Poorly-Graded Gravel, Gravel-Sand Mixture
		Gravels with Appreciable Fines	GM	Silty Gravel, Gravel-Sand-Silt Mixture
			GC	Clayey-Gravel, Gravel-Sand-Clay Mixture
	Sand and Sandy Soils	Clean Sands Little or no Fines	SW	Well-Graded Sand, Gravelly Sand
			SP	Poorly-Graded Sand, Gravelly Sand
		Sands with Appreciable Fines	SM	Silty Sand, Sand-Silt Mixture
			SC	Clayey-Sand, Sand-Clay Mixture
Fine-Grained Soils (More than 50% Smaller than No. 200 Sieve Size)	Silts and Clays	Liquid Limit Less Than 50	ML	Silt, Sandy Silt, Clayey Silt, Slight Plasticity
			CL	Lean Clay, Sandy Clay, Silty Clay, Low to Medium Plasticity
			OL	Organic Silts or Lean Clays, Low Plasticity
	Silts and Clays	Liquid Limit Greater Than 50	MH	Silt, High Plasticity
			CH	Fat Clay, High Plasticity
			OH	Organic Clay, Medium to High Plasticity
	Highly Organic Soils		PT	Peat, Humus, Swamp Soil

STRENGTH OF COHESIVE SOILS

Consistency	*Undrained Shear Strength (tsf)*	*Unconfined Comp. Strength (tsf)*
Very Soft	less than 0.125	less then 0.25
Soft	0.125 to 0.25	0.25 to 0.5
Medium Stiff	0.25 to 0.5	0.5 to 1.0
Stiff	0.5 to 1.0	1.0 to 2.0
Very Stiff	1.0 to 2.0	2.0 to 3.0
Hard	greater than 2.0	greater than 4.0

DENSITY OF GRANULAR SOILS

Descriptive Term	*Approximate N_{60}-Value Range*
Very Loose	0 to 4
Loose	5 to 10
Medium Dense	11 to 30
Dense	31 to 50
Very Dense	>50

N-Value (Blow Count) is the last two, 6-inch drive increments (i.e. 4/7/9, N = 7 + 9 = 16). Values are shown as a summation on the grid plot and shown in the Unit Dry Weight/SPT column.

RELATIVE COMPOSITION

Trace	0 to 10%
Little	10 to 20%
Some	20 to 35%
And	35 to 50%

OTHER TERMS

Layer - Inclusion greater than 3 inches thick.

Seam - Inclusion 1/8-inch to 3 inches thick

Parting - Inclusion less than 1/8-inch thick

Pocket - Inclusion of material that is smaller than sample diameter

UES

Relative composition and Unified Soil Classification System (USCS) designations are based on visual descriptions and are approximate only. If laboratory tests were performed to classify the soil, the USCS designation is shown in parenthesis.

APPENDIX C

Summary of Laboratory Results and
Laboratory Test Computation Sheets

Borehole	Depth	Water Content (%)	Dry Density (pcf)	Liquid Limit	Plastic Limit	Plasticity Index	%<#200 Sieve	Class-ification	Qu (tsf)	Strain %	Pocket Pen. (tsf)
B-1	1.0	19.0		49	19	30					
B-1	3.0	19.4	103.0						2.440	6.5	
B-1	6.0	21.6									
B-1	8.5	17.7									
B-1	13.5	17.8									
B-2	3.5	26.8									
B-2	6.0	18.5									
B-2	8.0	19.6	103.0	38	19	19			1.050	4.9	
B-2	13.5	18.9									
B-3	1.8	15.9		31	16	15					
B-3	3.5	14.1									
B-3	8.5	15.4									
B-3	14.5	21.9									

GEOTECH SUMMARY J045885.01 CSO COMBINED SEWER PROJECT.GPJ US_LAB.GDT 1/16/25

UES

Summary of Laboratory Results

Combined Sewer Overflow Separation
Atchison, Kansas
J045885.01

Water Content (ASTM D 2216 & D 4643)

GEOTECHNOLOGY INC
FROM THE GROUND UP

DATE 12/11/24 | PROJECT NAME Combined Sewer Project | PROJECT NO. J045885.01

General Test Information: Balance Used: BA- G00117 Oven Used: OV- G00101 Sheet 1 of 1

Boring No.	Sample No.	Depth (ft)	Water Content Information: Tare No.	Tare + Wet Soil (g)	Tare + Dry Soil (g)	Tare Weight (g)	Water Content (%)	Visual Identification (Color, Group Name & Symbol)
B-1	SS-1	1 / 2.5	WW-1	66.58	59.30	21.08		
	SS-3	6 / 7.5	WW-99	62.98	55.44	20.55		
	SS-4	8.5 / 10	M-4	52.72	48.01	21.34		
	SS-5	13.5 / 15	A-4	67.77	60.69	21.02		
B-2	SS-2	3.5 / 5	M-25	58.54	50.63	21.12		
	SS-3	6 / 7.5	M-43	64.10	57.37	21.03		
	SS-5	13.5 / 15	P-95	61.26	54.79	20.56		
B-3	SS-1	1.75 / 3.25	M-37	64.83	58.82	20.92		
	SS-2	3.5 / 5	M-20	57.79	53.23	20.99		
	SS-5	8.5 / 10	A-24	72.15	65.87	21.10		
	SS-6	13.5 / 15	A-9	66.30	58.16	21.07		

REMARKS:

Tested By: KEB Date: 12/11/24

Calc. By: ADC Date: 12/12/24

Checked By: MCL Date: 1/10/25

LIQUID LIMIT, PLASTIC LIMIT, AND PLASTICITY INDEX (ASTM D 4318)

GEOTECHNOLOGY INC
FROM THE GROUND UP

DATE	PROJECT NAME	Combined Sewer	PROJECT NO.	J045885.01
BORING NO.	B-1	SAMPLE NO.	SS-1	DEPTH (ft)	1'-2.5'

Sheet 1 of 1

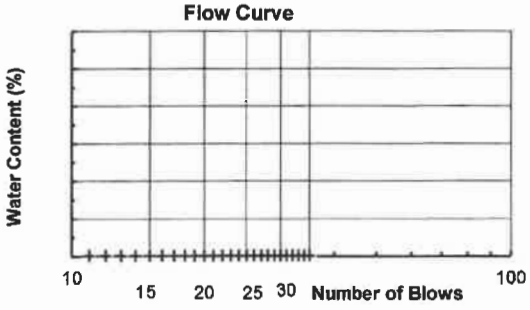
TESTING EQUIPMENT USED

Balance Used: BA-	Oven Used: OV- ✓		Liquid Limit:	Manual	✓
Plastic Limit:	Hand Rolled	✓	Apparatus No. (GOO) 202	Mechanical	
	Between Glass Plates		Casagrande / ASTM Grooving Tool. #1066	Metal	
	Mechanical Rolling Device			Plastic	✓

Visual Identification:

Signs of organic soil behavior: [ ] No; [ ] Yes (spongy PL; signs of oxidation; organic fibers; black color, humus odor)

SPECIMEN PREPARATION			
Wet		Washed on # 40 Sieve	
Dry (Air)		Dry Sieved on # 40 Sieve	
Dry (Oven)		Mechanically Pushed Through # 40 Sieve	
Mixed on Glass Plate and Removed Medium Plus Sand Particles			
Mixing water:	Distilled;	Demineralized; or other	



AS-RECEIVED WATER CONTENT (OVEN DRIED)

Container No.			
Mass Moist Soil + Container, M1 (g)			
Mass Dry Soil + Container, M2 (g)			
Mass Container, M3 (g)			
WATER CONTENT, w_o (%)			

Circle Approximate Max. Grain Size in "Sample": 3" 1-1/2" 3/4" 3/8" 3/8" #4 #10 <#10

PLASTIC LIMIT

Container No.	P-88	P-95		
Mass Moist Soil + Container, M1 (g)	30.34	30.34		
Mass Dry Soil + Container, M2 (g)	28.83	28.72		
Mass Container, M3 (g)	20.99	20.51		Average
WATER CONTENT, w (%)	19.3	19.7		19

LIQUID LIMIT

Container No.	M-41	A-24			
Mass Moist Soil + Container, M1 (g)	30.56	32.46	27.52		LINEAR REGRESSION ANALYSIS
Mass Dry Soil + Container, M2 (g)	27.92	~~28.05~~	26.67		w at N = 25 Blows
Mass Container, M3 (g)	21.28	21.16			Coef. of Determination, r^2
WATER CONTENT, w (%)					
NUMBER OF BLOWS, N	25	25			Average
LIQUID LIMIT, ASTM SINGLE POINT	48.7	48.1			49

Recommended range of Blow Count for Multiple Point Method A :
15 to 25, 20 to 30, and 25 to 35
Recommended range of Blow Count for Single Point Method B:

w_o or w = ((M1-M2) / (M2-M3)) * 100
LL = Water Content at N=25 blows, from Flow Curve
LL by Single Point = w * (N / 25) $^{0.121}$
PI = LL - PL
LI = (w_o - PL) / (LL - PL)

NOTES: (1) USCS: Unified Soil Classification System.
(2) Wet = Wet Preparation Method, and Dry = Dry Preparation Method

SUMMARY

TEST METHOD (2) A [ ] B [ ] Wet [ ] Dry [ ]	
AS-RECEIVED WATER CONTENT, w_o (%)	49
LIQUID LIMIT, LL	
PLASTIC LIMIT, PL	19
PLASTICITY INDEX, PI	
LIQUIDITY INDEX, LI	30
PERCENTAGE POINTS ABOVE/BELOW A-LINE	
PLASTICITY CHART CLASSIFICATION	

REMARKS:

Prepared By: KEB Date: 1/9/25
Tested By: WJW Date: 1/13/25
Calc. By: ADC Date: 1/14/25
Checked By: _____ Date: _____ MJL 1/10/25

LIQUID LIMIT, PLASTIC LIMIT, AND PLASTICITY INDEX (ASTM D 4318)

GEOTECHNOLOGY INC
FROM THE GROUND UP

DATE	PROJECT NAME	PROJECT NO.
	Combined Sewer	J045885.01

BORING NO.	SAMPLE NO.	DEPTH (ft)
B-2	ST-4	8'-10'

Sheet 1 of 1

TESTING EQUIPMENT USED

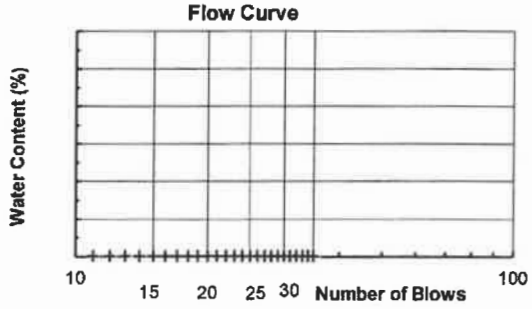
Balance Used: BA-GD117 Oven Used: OV-GD0101

Plastic Limit:			Liquid Limit:	
	Hand Rolled	X	Apparatus No. GD0123	Manual X
	Between Glass Plates		Casagrande / ASTM	Mechanical
	Mechanical Rolling Device		Grooving Tool: #67	Metal
				Plastic X

Visual Identification:

Signs of organic soil behavior: [ ] No; [ ] Yes (spongy PL; signs of oxidation; organic fibers; black color, humus odor)

SPECIMEN PREPARATION			
Wet		Washed on # 40 Sieve	
Dry (Air)		Dry Sieved on # 40 Sieve	
Dry (Oven)		Mechanically Pushed Through # 40 Sieve	
Mixed on Glass Plate and Removed Medium Plus Sand Particles			
Mixing water:		Distilled;	Demineralized; or other



AS-RECEIVED WATER CONTENT (OVEN DRIED)

Container No.			
Mass Moist Soil + Container, M1 (g)			
Mass Dry Soil + Container, M2 (g)			
Mass Container, M3 (g)			
WATER CONTENT, w_o (%)			

Circle Approximate Max. Grain Size in "Sample": 3" 1-1/2" 3/4" 3/8" 3/8" #4 #10 <#10

PLASTIC LIMIT

Container No.	WW-9	M-32			
Mass Moist Soil + Container, M1 (g)	30.59	31.49			
Mass Dry Soil + Container, M2 (g)	28.96	29.87			
Mass Container, M3 (g)	20.71	21.31		Average	
WATER CONTENT, w (%)	19.6	18.9		19	

LIQUID LIMIT

Container No.	WW-17	WW-12			
Mass Moist Soil + Container, M1 (g)	33.53	33.20			
Mass Dry Soil + Container, M2 (g)	30.69	29.87			
Mass Container, M3 (g)	21.07	21.11			
WATER CONTENT, w (%)					
NUMBER OF BLOWS, N	25	23			Average
LIQUID LIMIT, ASTM SINGLE POINT	38.1	37.6			38

LINEAR REGRESSION ANALYSIS	
w at N = 25 Blows	
Coef. of Determination, r^2	

Recommended range of Blow Count for Multiple Point Method A
15 to 25, 20 to 30, and 25 to 35
Recommended range of Blow Count for Single Point Method B:

w_o or w = ((M1-M2) / (M2-M3)) * 100
LL = Water Content at N=25 blows, from Flow Curve
LL by Single Point = w * (N / 25) $^{0.121}$
PI = LL - PL
LI = (w_o - PL) / (LL - PL)

NOTES: (1) USCS: Unified Soil Classification System.
(2) Wet = Wet Preparation Method, and Dry = Dry Preparation Method

SUMMARY

TEST METHOD (2) A [ ] B [ ] Wet [ ] Dry [ ]	
AS-RECEIVED WATER CONTENT, w_o (%)	
LIQUID LIMIT, LL	38
PLASTIC LIMIT, PL	19
PLASTICITY INDEX, PI	19
LIQUIDITY INDEX, LI	
PERCENTAGE POINTS ABOVE/BELOW A-LINE	
PLASTICITY CHART CLASSIFICATION	

REMARKS:

Prepared By: KEB Date: 1/8/25
Tested By: KEB Date: 1/8/25
Calc. By: ADC Date: 1/9/25
Checked By: MH Date: 1/10/25

LIQUID LIMIT, PLASTIC LIMIT, AND PLASTICITY INDEX (ASTM D 4318)

GEOTECHNOLOGY INC
FROM THE GROUND UP

DATE	PROJECT NAME	Combined Sewer	PROJECT NO.	J045885.01
BORING NO.	B-3	SAMPLE NO.	SS-1	DEPTH (ft)	1.75' - 3.25'

Sheet 1 of 1

TESTING EQUIPMENT USED

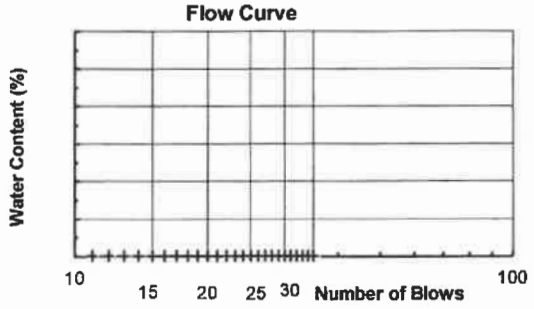
Balance Used: BA- GX0117 Oven Used: OV- G00101

Plastic Limit:	Hand Rolled	X	Liquid Limit: Apparatus No. G00123	Manual	X
	Between Glass Plates			Mechanical	
	Mechanical Rolling Device		Casagrande / ASTM Grooving Tool: #67	Metal	
				Plastic	X

Visual Identification:

Signs of organic soil behavior: [ ] No; [ ] Yes (spongy PL; signs of oxidation; organic fibers; black color, humus odor)

SPECIMEN PREPARATION			
Wet		Washed on # 40 Sieve	
Dry (Air)		Dry Sieved on # 40 Sieve	
Dry (Oven)		Mechanically Pushed Through # 40 Sieve	
Mixed on Glass Plate and Removed Medium Plus Sand Particles			
Mixing water:	Distilled;	Demineralized; or other	



AS-RECEIVED WATER CONTENT (OVEN DRIED)

Container No.			
Mass Moist Soil + Container, M1 (g)			
Mass Dry Soil + Container, M2 (g)			
Mass Container, M3 (g)			
WATER CONTENT, w_o (%)			

Circle Approximate Max. Grain Size in "Sample": 3" 1-1/2" 3/4" 3/8" #4 #10 <#10

PLASTIC LIMIT

Container No.	A-9	WW-17		
Mass Moist Soil + Container, M1 (g)	30.58	30.86		
Mass Dry Soil + Container, M2 (g)	29.33	29.50		
Mass Container, M3 (g)	21.01	21.05		Average
WATER CONTENT, w (%)	15.1	16.1		16

LIQUID LIMIT

Container No.	M-26	M-32			
Mass Moist Soil + Container, M1 (g)	32.33	32.82			
Mass Dry Soil + Container, M2 (g)	29.63	30.12			
Mass Container, M3 (g)	20.93	21.29			
WATER CONTENT, w (%)					
NUMBER OF BLOWS, N	26	24			Average
LIQUID LIMIT, ASTM SINGLE POINT	31.2	30.4			31

LINEAR REGRESSION ANALYSIS	
w at N = 25 Blows	
Coef. of Determination, r^2	

Recommended range of Blow Count for Multiple Point Method A
15 to 25, 20 to 30, and 25 to 35.
Recommended range of Blow Count for Single Point Method B:

w_o or w = ((M1-M2) / (M2-M3)) * 100
LL = Water Content at N=25 blows, from Flow Curve
LL by Single Point = w * (N / 25) $^{0.121}$
PI = LL - PL
LI = (w_o - PL) / (LL - PL)

NOTES: (1) USCS: Unified Soil Classification System.
(2) Wet = Wet Preparation Method, and Dry = Dry Preparation Method

REMARKS:

SUMMARY

TEST METHOD (2) A [ ] B [ ] Wet [ ] Dry [ ]	
AS-RECEIVED WATER CONTENT, w_o (%)	
LIQUID LIMIT, LL	31
PLASTIC LIMIT, PL	
PLASTICITY INDEX, PI	16
LIQUIDITY INDEX, LI	
PERCENTAGE POINTS ABOVE/BELOW A-LINE	15
PLASTICITY CHART CLASSIFICATION	

Prepared By: KEB Date: 1/13/25
Tested By: KEB Date: 1/13/25
Calc. By: ADC Date: 1/14/25
Checked By: MLQ Date: 1/16/25

1C

GEOTECHNOLOGY INC
FROM THE GROUND UP

LABORATORY LOG OF TUBE SAMPLE

Project No.: J045885.01 Project Name.: Combined Sewer Date Opened: 12/11/24 By: KEB

Boring No.: B-1 Sample No.: ST-2 Depth (ft): 3' to 5'

Type [X] Std. "Shelby" [ ] Other: _____

Tube Dimensions: Outside Diameter (in): 3 Recovery (in): 24

Tube Condition:	Dented	End(s) not sealed	Other:

Sample Condition	Good	Fair	Poor	Disturbed

Remarks: _____

Tube Scale (in)	Sample Use (Test Type)	Soil Identification and Processing Remarks (Draw lines to indicate top and botton of soil's surface and where sample was taken. Identify cut lines with: "Cut" "Date: m/d/y" "Your Initials")
0.0		Clay - light yellowish brown, stiff, fat, trace sand/rock
2.0		
4.0		
6.0		
8.0		
10.0		
12.0		
14.0		
16.0		
18.0	UC	
20.0		
22.0		
24.0		

Natural Moisture Content				
Tare Number	Wet Mass + Tare (g)	Dry Mass+ Tare (g)	Tare Mass (g)	Water Content (%)

Tested By: KEB Date: 12/11/24

Calculated By: ACC Date: 12/12/24

Checked By: MA Date: 1/10/25

UNCONFINED COMPRESSION TEST / ASTM D2166
AND UNIT WEIGHT DETERMINATION / ASTM 7263

Job No.: J045885.01
Job Name: Combined Sewer
Date: 12/11/24

Boring No. B-1 Sample No. ST-2
Depth 3'-5' Technician KEB
Balance Used G00111
Oven Used G00102

V

INITIAL CONDITION OF SPECIMEN			
Diameter	2.864, 2.868, 2.861 (in.)		(mm)
Height	5.664, 5.655, 5.645 (in.)		(mm)
Area		(in2.)	(cm2)
Volume		(ft3.)	(cm3)
Wet Weight		(lbs)	1264.6 (g)
Wet Density	123	(pcf)	(kg/cm3)
Dry Density	~~117~~ 103	(pcf)	(kg/cm3)

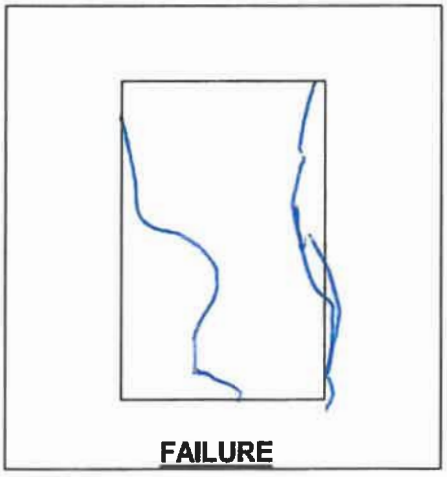
WATER CONTENT	
Tare Number	X-160
Wet Weight + Tare	129.0
Dry Weight + Tare	109.3
Weight of Water	
Tare Weight	7.6
Weight of Dry Soil	
Water Content (%)	~~17.8~~ 19.4

☐ Load Cell

Dial Reading (0.001")	Load ☐ (lb) ☐(Div)	Dial Reading (0.001")	Load ☐ (lb) ☐(Div)
0		520	
20		40	
40		60	
60		80	
80		600	
100		20	
20		40	
40		60	
60		80	
80		700	
200		20	
20		40	
40		60	
60		80	
80		800	
300		20	
20		40	
40		60	
60		80	
80		900	
400		20	
20		40	
40		60	
60		80	
80		1,000	
500			

FAILURE CONDITION

Deformation Rate (in./min.)
Failure Strain (%) 6.51
Compression Strength qu (tsf) 2.44
Qu/2 (tsf) 1.22



FAILURE

Remarks:

Recovery
P.P.

MLQ 1/16/25

UC
AL

V

GEOTECHNOLOGY INC
FROM THE GROUND UP

LABORATORY LOG OF TUBE SAMPLE

Project No.: J045885.01 Project Name.: Combined Sewer Date Opened: 12/11/24 By: KEB

Boring No.: B-2 Sample No.: ST-4 Depth (ft): 8' to 10'

Type [X] Std. "Shelby" [ ] Other: _____

Tube Dimensions: Outside Diameter (in): 3 Recovery (in): 24

Tube Condition:	Dented	End(s) not sealed	Other:

Sample Condition	Good	Fair	Poor	Disturbed

Remarks: _____

Tube Scale (in)	Sample Use (Test Type)	Soil Identification and Processing Remarks (Draw lines to indicate top and botton of soil's surface and where sample was taken. Identify cut lines with: "Cut" "Date: m/d/y" "Your Initials")
0.0		Slough
2.0		Clay - dark reddish brown, lean, some organic material
4.0		
6.0		
8.0		
10.0		
12.0		
14.0		
16.0		
18.0	UC	
20.0		
22.0		
24.0		

Natural Moisture Content				
Tare Number	Wet Mass + Tare (g)	Dry Mass+ Tare (g)	Tare Mass (g)	Water Content (%)

Tested By: KEB Date: 12/11/24

Calculated By: ADC Date: 12/12/24

Checked By: _____ Date: _____

M.Q 1/10/25

UNCONFINED COMPRESSION TEST / ASTM D2166
AND UNIT WEIGHT DETERMINATION / ASTM 7263

Job No.: J045885.01
Job Name: Combined Sewer
Date: 12/11/24

Boring No. B-2 Sample No. ST-4
Depth 8'-10' Technician KEB
Balance Used G00111
Oven Used G00101

INITIAL CONDITION OF SPECIMEN				
Diameter	2.866, 2.872, 2.861	(in.)		(mm)
Height	5.653, 5.632, 5.655	(in.)		(mm)
Area		(in2.)		(cm2)
Volume		(ft3.)		(cm3)
Wet Weight		(lbs)	1176.9	(g)
Wet Density	123	(pcf)		(kg/cm3)
Dry Density	103	(pcf)		(kg/cm3)

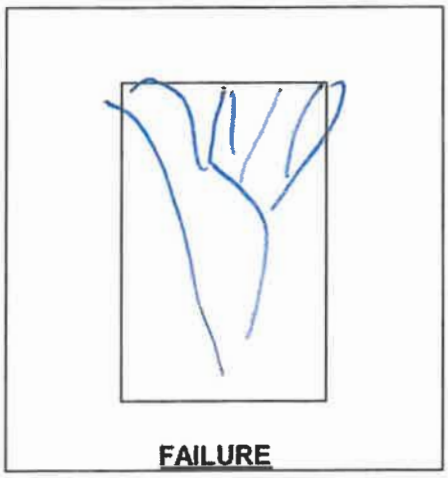
WATER CONTENT	
Tare Number	X-107
Wet Weight + Tare	146.1
Dry Weight + Tare	123.4
Weight of Water	
Tare Weight	7.7
Weight of Dry Soil	
Water Content (%)	19.6

☐ Load Cell

Dial Reading (0.001")	Load ☐ (lb) ☐(Div)	Dial Reading (0.001")	Load ☐ (lb) ☐(Div)
0		520	
20		40	
40		60	
60		80	
80		600	
100		20	
20		40	
40		60	
60		80	
80		700	
200		20	
20		40	
40		60	
60		80	
80		800	
300		20	
20		40	
40		60	
60		80	
80		900	
400		20	
20		40	
40		60	
60		80	
80		1,000	
500			

FAILURE CONDITION

Deformation Rate (in./min.) _____
Failure Strain (%) 4.91
Compression Strength qu (tsf) 1.05
Qu/2 (tsf) 0.52



FAILURE

Remarks: _____

Recovery _____
P.P. _____

1/16/25

APPENDIX D

Corrosion Potential Test Results

Pace Analytical Services, LLC
9608 Loiret Blvd.
Lenexa, KS 66219
(913)599-5665

December 24, 2024

Matt McQuality
GEOTECHNOLOGY, INC.
5055 Antioch Road
Shawnee, KS 66203

RE: Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

Dear Matt McQuality:

Enclosed are the analytical results for sample(s) received by the laboratory on December 05, 2024. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Kansas City

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Angie Brown
angie.brown@pacelabs.com
1(913)563-1402
Project Manager

Enclosures

cc: Angie Coates, GEOTECHNOLOGY, INC.

REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

Pace Analytical Services Kansas
9608 Loiret Boulevard, Lenexa, KS 66219
Arkansas Certification #: 88-00679
Illinois Certification #: 2000302023-6
Colorado Division of Oil and Public Safety
Iowa Certification #: 118
Kansas Field Laboratory Certification #: E-92587
Kansas/NELAP Certification #: E-10116
Louisiana Certification #: 03055
Missouri Inorganic Drinking Water Certification
Nevada Certification #: KS000212024-1
Oklahoma Certification #: 2023-073
Texas Certification #: T104704407-23-17
Utah Certification #: KS000212022-13

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SAMPLE SUMMARY

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

Lab ID	Sample ID	Matrix	Date Collected	Date Received
60465963001	**B-2 10'-10.5'**	Solid	12/05/24 09:00	12/05/24 12:20

REPORT OF LABORATORY ANALYSIS

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SAMPLE ANALYTE COUNT

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
60465963001	**B-2 10'-10.5'**	EPA 9045	TML	1	PASI-K
		SM 2580B	SR1	1	PASI-K
		EPA 120.1 Resistivity	SR1	1	PASI-K
		EPA 9056	AAA	2	PASI-K

PASI-K = Pace Analytical Services - Kansas City

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PROJECT NARRATIVE

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

Date: December 24, 2024

The analysis results for Sulfide are located as a sublab file following the chain of custody section of the report.

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ANALYTICAL RESULTS

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

Sample: B-2 10'-10.5' **Lab ID: 60465963001** Collected: 12/05/24 09:00 Received: 12/05/24 12:20 Matrix: Solid

Results reported on a "wet-weight" basis

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
9045 pH Soil	Analytical Method: EPA 9045 Pace Analytical Services - Kansas City							
pH at 25 Degrees C	**5.8**	Std. Units	0.10	1		12/18/24 17:45		
Oxidation/Reduction Potential	Analytical Method: SM 2580B Pace Analytical Services - Kansas City							
Oxidation/Reduction Potential	**231.9**	mV	1.0	1		12/12/24 11:34		H1
Resistivity	Analytical Method: EPA 120.1 Resistivity Pace Analytical Services - Kansas City							
Resistivity	**225**	ohms-cm	100	1		12/17/24 15:15		
9056 IC Anions	Analytical Method: EPA 9056 Preparation Method: EPA 9056 Pace Analytical Services - Kansas City							
Chloride	**507**	mg/kg	93.8	10	12/11/24 12:00	12/11/24 19:42	16887-00-6	
Sulfate	**476**	mg/kg	93.8	10	12/11/24 12:00	12/11/24 19:42	14808-79-8	

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

QC Batch: 919727
QC Batch Method: EPA 9045
Analysis Method: EPA 9045
Analysis Description: 9045 pH
Laboratory: Pace Analytical Services - Kansas City
Associated Lab Samples: 60465963001

SAMPLE DUPLICATE: 3642974

Parameter	Units	60465629001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	6.2	6.3	2	3	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

QC Batch: 918998
QC Batch Method: SM 2580B
Analysis Method: SM 2580B
Analysis Description: Oxidation/Reduction Potential
Laboratory: Pace Analytical Services - Kansas City

Associated Lab Samples: 60465963001

SAMPLE DUPLICATE: 3639669

Parameter	Units	20338781001 Result	Dup Result	RPD	Max RPD	Qualifiers
Oxidation/Reduction Potential	mV	271.6	271.0			H3

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

QC Batch: 919086
QC Batch Method: EPA 9056
Analysis Method: EPA 9056
Analysis Description: 9056 IC Anions
Laboratory: Pace Analytical Services - Kansas City
Associated Lab Samples: 60465963001

METHOD BLANK: 3639971 Matrix: Solid
Associated Lab Samples: 60465963001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	ND	100	12/11/24 17:34	
Sulfate	mg/kg	ND	100	12/11/24 17:34	

LABORATORY CONTROL SAMPLE: 3639972

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	500	473	95	80-120	
Sulfate	mg/kg	500	470	94	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3639973 3639974

Parameter	Units	60465733001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	20800	2620	2650	17300	18500	-134	-89	80-120	7	15	M1
Sulfate	mg/kg	39800	2620	2650	33300	38100	-248	-65	80-120	13	15	M1

SAMPLE DUPLICATE: 3639975

Parameter	Units	60465733001 Result	Dup Result	RPD	Max RPD	Qualifiers
Chloride	mg/kg	20800	ND		15	
Sulfate	mg/kg	39800	ND		15	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALIFIERS

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

H1	Analysis conducted outside the EPA method holding time.
H3	Sample was received or analysis requested beyond the recognized method holding time.
M1	Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: COMBINED SEWER/J045885.01
Pace Project No.: 60465963

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
60465963001	**B-2 10'-10.5'**	EPA 9045	919727		
60465963001	**B-2 10'-10.5'**	SM 2580B	918998		
60465963001	**B-2 10'-10.5'**	EPA 120.1 Resistivity	919714		
60465963001	**B-2 10'-10.5'**	EPA 9056	919086	EPA 9056	919457

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WO# : 60465963

60465963

Pace ANALYTICAL SERVICES	DC#_Title: ENV-FRM-LENE-0009_Sampl		
	Revision: 2	Effective Date: 01/12/2022	Issued By: Lenexa

Client Name: Geotech

Courier: FedEx ☐ UPS ☐ VIA ☐ Clay ☐ PEX ☐ ECI ☐ Pace ☐ Xroads ☐ Client ☑ Other ☐

Tracking #: _____ Pace Shipping Label Used? Yes ☐ No ☑

Custody Seal on Cooler/Box Present: Yes ☐ No ☑ Seals intact: Yes ☐ No ☑

Packing Material: Bubble Wrap ☐ Bubble Bags ☐ Foam ☐ None ☑ Other ☐

Thermometer Used: T298 **Type of Ice:** Wet Blue (None)

Cooler Temperature (°C): As-read 1.7 Corr. Factor -0.1 Corrected 1.6

Date and initials of person examining contents: MM 12/5/24

Temperature should be above freezing to 6°C

Chain of Custody present:	☑Yes ☐No ☐N/A	
Chain of Custody relinquished:	☑Yes ☐No ☐N/A	
Samples arrived within holding time:	☑Yes ☐No ☐N/A	
Short Hold Time analyses (<72hr):	☐Yes ☑No ☐N/A	
Rush Turn Around Time requested:	☐Yes ☑No ☐N/A	
Sufficient volume:	☑Yes ☐No ☐N/A	
Correct containers used:	☑Yes ☐No ☐N/A	
Pace containers used:	☐Yes ☑No ☐N/A	
Containers intact:	☑Yes ☐No ☐N/A	
Unpreserved 5035A / TX1005/1006 soils frozen in 48hrs?	☐Yes ☑No ☐N/A	
Filtered volume received for dissolved tests?	☐Yes ☑No ☐N/A	
Sample labels match COC: Date / time / ID / analyses	☑Yes ☐No ☐N/A	
Samples contain multiple phases? Matrix: SL	☐Yes ☑No ☐N/A	
Containers requiring pH preservation in compliance? (HNO_3, H_2SO_4, HCl<2; NaOH>9 Sulfide, NaOH>10 Cyanide) (Exceptions: VOA, Micro, O&G, KS TPH, OK-DRO) LOT#:	☐Yes ☐No ☑N/A	List sample IDs, volumes, lot #'s of preservative and the date/time added.
Cyanide water sample checks: Lead acetate strip turns dark? (Record only)	☐Yes ☐No	
Potassium iodide test strip turns blue/purple? (Preserve)	☐Yes ☐No	
Trip Blank present:	☐Yes ☐No ☑N/A	
Headspace in VOA vials (>6mm):	☐Yes ☐No ☑N/A	
Samples from USDA Regulated Area: State:	☐Yes ☑No ☐N/A	
Additional labels attached to 5035A / TX1005 vials in the field?	☐Yes ☐No ☑N/A	

Client Notification/ Resolution: Copy COC to Client? Y / N Field Data Required? Y / N

Person Contacted: _____ Date/Time: _____

Comments/ Resolution: _____

Project Manager Review: _____ Date: _____

Qualtrax Document ID: 30468

CHAIN-OF-CUSTODY Analytical Request Document

Pace Analytical

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevent fields

LAB USE ONLY- Affix Workorder/Login Label Here or List Pace Workorder Number or MTJL Log-in Number Here: 60465963

ALL SHADED AREAS are for LAB USE ONLY

Company: Geotechnology, LLC / dba UES	Billing Information:
Address: 5055 Antioch Road Shawnee KS 66203	
Report To: Matt McQuality	Email To:
Copy To:	Site Collection Info/Address:
Customer Project Name/Number: Combined Sewer / J045785.01	State: / County/City: Time Zone Collected: [ ] PT [ ] MT [ ] CT [ ] ET
Phone: 913-375-4389 Email: mmcquality@teamues.com	Site/Facility ID #: Compliance Monitoring? [ ] Yes [ ] No
Collected By (print): Ryan White	Purchase Order #: Quote #: DW PWS ID #: DW Location Code:
Collected By (signature):	Turnaround Date Required: Immediately Packed on Ice: [ ] Yes [ ] No
Sample Disposal: [ ] Dispose as appropriate [ ] Return [ ] Archive: [ ] Hold:	Rush: [ ] Same Day [ ] Next Day [ ] 2 Day [ ] 3 Day [ ] 4 Day [ ] 5 Day (Expedite Charges Apply)
	Field Filtered (if applicable): [ ] Yes [ ] No Analysis:

Container Preservative Type **

Lab Project Manager:

** Preservative Types: (1) nitric acid, (2) sulfuric acid, (3) hydrochloric acid, (4) sodium hydroxide, (5) zinc acetate, (6) methanol, (7) sodium bisulfate, (8) sodium thiosulfate, (9) hexane, (A) ascorbic acid, (B) ammonium sulfate, (C) ammonium hydroxide, (D) TSP, (U) Unpreserved, (O) Other

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Wastewater (WW), Product (P), Soil/Solid (SL), Oil (OL), Wipe (WP), Air (AR), Tissue (TS), Bioassay (B), Vapor (V), Other (OT)

Customer Sample ID	Matrix *	Comp / Grab	Collected (or Composite Start) Date	Collected (or Composite Start) Time	Composite End Date	Composite End Time	Res Cl	# of Ctns	Oxidation-reduction potential	Soil resistivity	pH	Chloride ions	Sulfide ions	Soluble sulfates	Lab Profile/Line.
B-2 10'-10.5'			12/5	9:00am											

Lab Sample Receipt Checklist:

Custody Seals Present/Intact	Y	N	NA
Custody Signatures Present	Y	N	NA
Collector Signature Present	Y	N	NA
Bottles Intact	Y	N	NA
Correct Bottles	Y	N	NA
Sufficient Volume	Y	N	NA
Samples Received on Ice	Y	N	NA
VOA - Headspace Acceptable	Y	N	NA
USDA Regulated Soils	Y	N	NA
Samples in Holding Time	Y	N	NA
Residual Chlorine Present	Y	N	NA
Cl Strips:			
Sample pH Acceptable	Y	N	NA
pH Strips:			
Sulfide Present	Y	N	NA
Lead Acetate Strips:			

LAB USE ONLY:
Lab Sample # / Comments:

Customer Remarks / Special Conditions / Possible Hazards:

Type of Ice Used: Wet Blue Dry None

Packing Material Used:

Radchem sample(s) screened (<500 cpm): Y N NA

SHORT HOLDS PRESENT (<72 hours): Y N N/A

Lab Tracking #: 2919090

Samples received via: FEDEX UPS Client Courier Pace Courier

Lab Sample Temperature Info:
Temp Blank Received: Y N NA
Therm ID#:
Cooler 1 Temp Upon Receipt: ____oC
Cooler 1 Therm Corr. Factor: ____oC
Cooler 1 Corrected Temp: 1.6 oC
Comments:

Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:
	12/5/24/12:20pm		12/5 1220
Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:
Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:

MTJL LAB USE ONLY
Table #:
Acctnum:
Template:
Prelogin:
PM:
PB:

Trip Blank Received: Y N NA
HCL MeOH TSP Other

Non Conformance(s): YES / NO

Page: ____ of: ____

Client: Geotech Profile/EZ # 7516, 1

Site: Combined Sewer Notes

COC Line Item	Matrix	VG9H	DG9H	DG9Q	VG9U	DG9U	DG9M	DG9B	BG1U	AG1H	AG1U	AG2U	AG3S	AG4U	AG5U	JGFU	WGKU	WGDU	BP1U	BP2U	BP3U	BP1N	BP3N	BP3F	BP3S	BP3B	BP3Z	WPDU	ZPLC	Other		
1	SL																												1			
2																																
3																																
4																																
5																																
6																																
7																																
8																																
9																																
10																																
11																																
12																																

Container Codes

Glass			
DG9B	40mL bisulfate clear vial	WGKU	8oz clear soil jar
DG9H	40mL HCl amber voa vial	WGFU	4oz clear soil jar
DG9M	40mL MeOH clear vial	WG2U	2oz clear soil jar
DG9Q	40mL TSP amber vial	JGFU	4oz unpreserved amber wide
DG9S	40mL H2SO4 amber vial	AG0U	100mL unores amber glass
DG9T	40mL Na Thio amber vial	AG1H	1L HCl amber glass
DG9U	40mL amber unpreserved	AG1S	1L H2SO4 amber glass
VG9H	40mL HCl clear vial	AG1T	1L Na Thiosulfate clear/amber glass
VG9T	40mL Na Thio. clear vial	AG1U	1liter unpres amber glass
VG9U	40mL unpreserved clear vial	AG2N	500mL HNO3 amber glass
BG1S	1liter H2SO4 clear glass	AG2S	500mL H2SO4 amber glass
BG1U	1liter unpres glass	AG3S	250mL H2SO4 amber glass
BG3H	250mL HCL Clear glass	AG2U	500mL unpres amber glass
BG3U	250mL Unpres Clear glass	AG3U	250mL unpres amber glass
WGDU	16oz clear soil jar	AG4U	125mL unpres amber glass
		AG5U	100mL unpres amber glass

Plastic	
BP1B	1L NAOH plastic
BP1N	1L HNO3 plastic
BP1S	1L H2SO4 plastic
BP1U	1L unpreserved plastic
BP1Z	1L NaOH, Zn Acetate
BP2B	500mL NAOH plastic
BP2N	500mL HNO3 plastic
BP2S	500mL H2SO4 plastic
BP2U	500mL unpreserved plastic
BP2Z	500mL NaOH, Zn Acetate
BP3B	250mL NaOH plastic
BP3F	250mL HNO3 plastic - field filtered
BP3N	250mL HNO3 plastic
BP3U	250mL unpreserved plastic
BP3S	250mL H2SO4 plastic
BP3Z	250mL NaOH, Zn Acetate
BP4U	125mL unpreserved plastic
BP4N	125mL HNO3 plastic
BP4S	125mL H2SO4 plastic
WPDU	16oz unpresserved plstic

Misc.	
I	Wipe/Swab
SP5T	120mL Coliform Na Thiosulfate
ZPLC	Ziploc Bag
AF	Air Filter
C	Air Cassettes
R	Terracore Kit
U	Summa Can

Matrix	
WT	Water
SL	Solid
NAL	Non-aqueous Liquid
OL	OIL
WP	Wipe
DW	Drinking Water

Work Order Number: WO#: 60465963

PM: AKB Due Date: 12/19/24

CLIENT: GEOTECH_KC

B064

Internal Transfer Chain of Custody

Pace

☐ Rush Multiplier____X
☐ Samples Pre-Logged into eCOC

State Of Origin: KS
Cert. Needed: ☒ Yes ☐ No

Workorder: 60465963 **Workorder Name:** COMBINED SEWER/J045885.01

Owner Received Date: 12/5/2024 **Results Requested By:** 12/19/2024

Report To	Subcontract To
Angie Brown Pace Analytical Kansas 9608 Loiret Blvd. Lenexa, KS 66219 Phone 1(913)563-1402	Pace National 12065 Lebanon Rd Mt. Juliet, TN 37122 Phone (615) 758-5858

Item	Sample ID	Sample Type	Collect Date/Time	Lab ID	Matrix	Preserved Containers: Unpreserved	Sulfide	Requested Analysis	LAB USE ONLY
1	B-2 10'-10.5	PS	12/5/2024 09:00	60465963001	Solid	1	X		-01
2									
3									
4									
5									

Transfers	Released By	Date/Time	Received By	Date/Time
1	JAPAE	12/6/24 1100		
2				
3			Christopher M. Collins	12/6/24 0900

Comments: KS sample located 6091-R6-S2BS

PLEASE PULL TO SPLIT INTO JGFU

Cooler Temperature on Receipt °C | **Custody Seal** Y or N | **Received on Ice** Y or N | **Samples Intact** Y or N

***In order to maintain client confidentiality, location/name of the sampling site, sampler's name and signature may not be provided on this COC document. This chain of custody is considered complete as is since this information is available in the owner laboratory.

0.210=0.2

Sample Receipt Checklist

COC Seal Present/Intact	Y N	If Applicable	
COC Signed/Accurate	Y N	VOA Zero Headspace	Y N
Bottles arrive intact	Y N	Pres. Correct/Check	Y N
Correct Bottles Used	Y N		
Sufficient volume sent	Y N	1 TOTAL	
RA Screen <0.5 mR/hr	Y N		

LR606972

Pace

Ship To:
Pace National
12065 Lebanon Rd
Mt. Juliet, TN 37122
Phone (615) 758-5858

INTER_LABORATORY WORK ORDER # 60465963
(To be completed by sending lab)

Sending Project No:	60465963
Receiving Project No:	
Check Box for Consolidated Invoice:	☐
Date Prepared:	12/06/24
REQUESTED COMPLETION DATE:	**12/19/2024**

Sending Region	IR60-Kansas	Sending Project Mgr.	Angie Brown
Receiving Region	IR850-Pace National	External Client	Universal Engineering_dba Geotechnology Inc
State of Sample Origin	KS	QC Deliverable	STD REPORT

All questions should be addressed to sending project manager.

Requested Reportable Units _____ Report Wet or Dry Weight? Wet Cert. Needed YES

WORK REQUESTED

Method Description	Container Type	Quantity of containers	Preservative	Quantity of Samples	Acode	Acode Desc
Sulfide	JGFU	1	Unpreserved	1	SI-21WET	SUB PASI WET

Special Requirements: Pace Regulatory Report (870)

FOR ANALYTICAL WORK COMPLETED THIS SECTION ALSO

Return Samples to Sending Region: ☐ Yes ☒ No

DISPOSITION of FORM

Original sent to the receiving lab - Copy kept at the sending lab.

When work completed: Original sent to the ABM at the receiving laboratory. Copies are made to corporate as needed.

ANALYTICAL REPORT

December 10, 2024

Pace Analytical - Lenexa, KS

Sample Delivery Group:	L1806972
Samples Received:	12/07/2024
Project Number:	60465963
Description:	Comibined Sewer/J045885.01
Site:	001
Report To:	Angie Brown 9608 Loiret Boulevard Lenexa, KS 66219

Entire Report Reviewed By: Nancy F. McLain

Nancy McLain
Project Manager

Results relate only to the items tested or calibrated and are reported as rounded values. This test report shall not be reproduced, except in full, without written approval of the laboratory. Where applicable, sampling conducted by Pace Analytical National is performed per guidance provided in laboratory standard operating procedures ENV-SOP-MTJL-0067 and ENV-SOP-MTJL-0068. Where sampling conducted by the customer, results relate to the accuracy of the information provided, and as the samples are received.

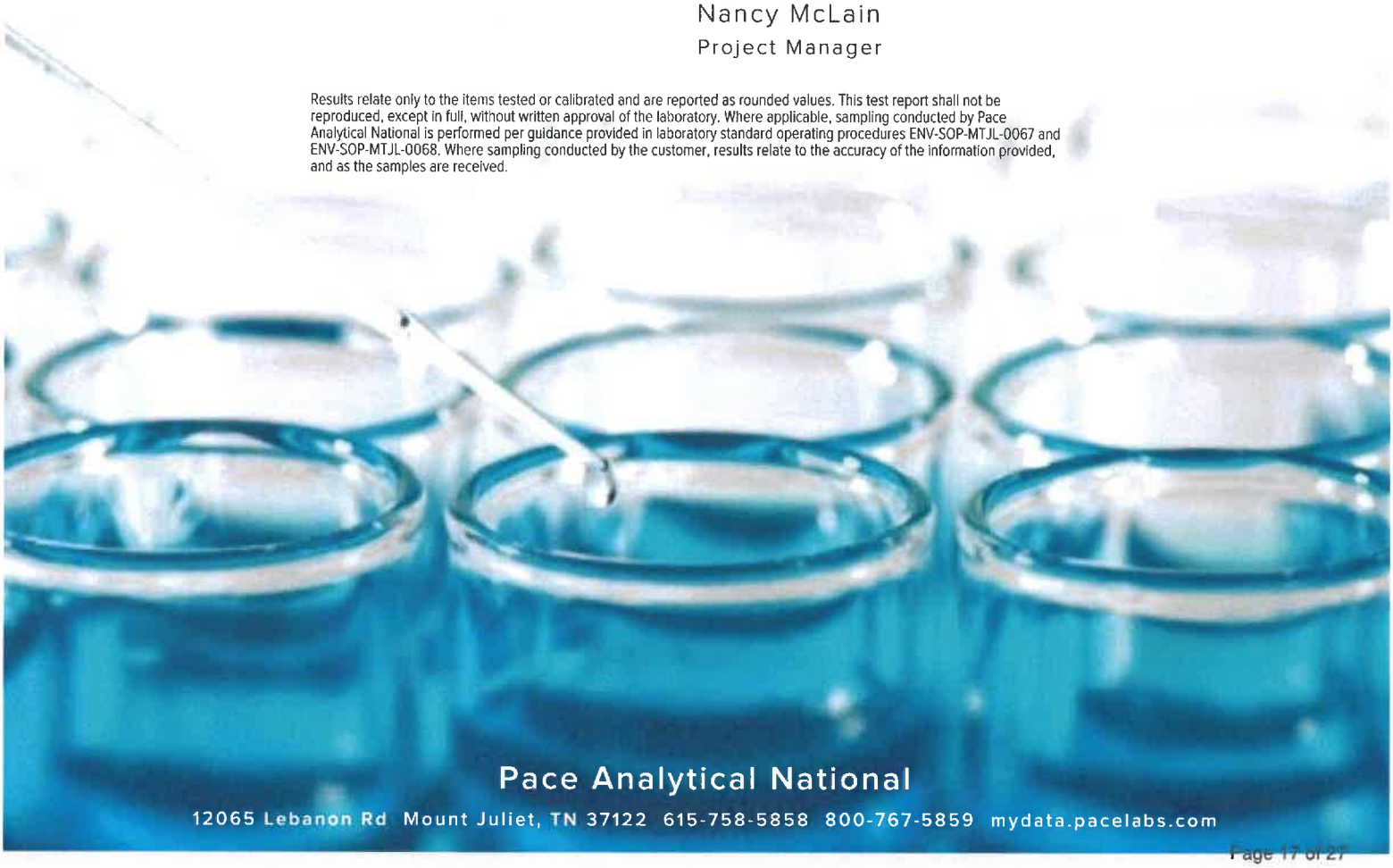


TABLE OF CONTENTS

Cp: Cover Page 1

Tc: Table of Contents 2

Ss: Sample Summary 3

Cn: Case Narrative 4

Sr: Sample Results 5

B-2 10'-10.5' L1806972-01 5

Qc: Quality Control Summary 6

Total Solids by Method 2540 G-2011 6

Wet Chemistry by Method 9034/9030B 7

Gl: Glossary of Terms 8

Al: Accreditations & Locations 9

Sc: Sample Chain of Custody 10

SAMPLE SUMMARY

	Collected by	Collected date/time	Received date/time
B-2 10'-10.5' L1806972-01 Solid		12/05/24 09:00	12/07/24 09:00

Method	Batch	Dilution	Preparation date/time	Analysis date/time	Analyst	Location
Total Solids by Method 2540 G-2011	WG2415309	1	12/09/24 14:17	12/09/24 14:26	CMB	Mt. Juliet, TN
Wet Chemistry by Method 9034/9030B	WG2413751	1	12/07/24 16:59	12/07/24 22:00	CJW	Mt. Juliet, TN

CASE NARRATIVE

All sample aliquots were received at the correct temperature, in the proper containers, with the appropriate preservatives, and within method specified holding times, unless qualified or notated within the report. Where applicable, all MDL (LOD) and RDL (LOQ) values reported for environmental samples have been corrected for the dilution factor used in the analysis. All Method and Batch Quality Control are within established criteria except where addressed in this case narrative, a non-conformance form or properly qualified within the sample results. By my digital signature below, I affirm to the best of my knowledge, all problems/anomalies observed by the laboratory as having the potential to affect the quality of the data have been identified by the laboratory, and no information or data have been knowingly withheld that would affect the quality of the data.

Nancy F. McLain

Nancy McLain
Project Manager

Total Solids by Method 2540 G-2011

Analyte	Result	Qualifier	Dilution	Analysis	Batch
	%			date / time	
Total Solids	83.8		1	12/09/2024 14:26	WG2415309

Wet Chemistry by Method 9034/9030B

Analyte	Result (dry)	Qualifier	MDL (dry)	RDL (dry)	Dilution	Analysis	Batch
	mg/kg		mg/kg	mg/kg		date / time	
Sulfide	U		33.5	89.5	1	12/07/2024 22:00	WG2413751

QUALITY CONTROL SUMMARY

L1806972-01

Method Blank (MB)

(MB) R4155742-1 12/09/24 14:26

Analyte	MB Result	MB Qualifier	MB MDL	MB RDL
	%		%	%
Total Solids	0.00300			

L1806952-30 Original Sample (OS) • Duplicate (DUP)

(OS) L1806952-30 12/09/24 14:26 • (DUP) R4155742-3 12/09/24 14:26

Analyte	Original Result	DUP Result	Dilution	DUP RPD	DUP Qualifier	DUP RPD Limits
	%	%		%		%
Total Solids	84.2	82.5	1	1.99		10

Laboratory Control Sample (LCS)

(LCS) R4155742-2 12/09/24 14:26

Analyte	Spike Amount	LCS Result	LCS Rec.	Rec. Limits	LCS Qualifier
	%	%	%	%	
Total Solids	50.0	50.0	99.9	90.0-110	

Method Blank (MB)

(MB) R4154972-1 12/07/24 22:00

Analyte	MB Result mg/kg	MB Qualifier	MB MDL mg/kg	MB RDL mg/kg
Sulfide	U		28.1	75.0

Laboratory Control Sample (LCS)

(LCS) R4154972-2 12/07/24 22:00

Analyte	Spike Amount mg/kg	LCS Result mg/kg	LCS Rec. %	Rec. Limits %	LCS Qualifier
Sulfide	100	93.6	93.6	53.8-124	

L1805572-01 Original Sample (OS) • Matrix Spike (MS) • Matrix Spike Duplicate (MSD)

(OS) L1805572-01 12/07/24 22:00 • (MS) R4154972-3 12/07/24 22:00 • (MSD) R4154972-4 12/07/24 22:00

Analyte	Spike Amount (dry) mg/kg	Original Result (dry)	MS Result (dry)	MSD Result (dry)	MS Rec. %	MSD Rec. %	Dilution	Rec. Limits %	MS Qualifier	MSD Qualifier	RPD %	RPD Limits %
Sulfide	107	39.8	93.7	97.2	50.3	53.6	1	10.0-136	J	J	3.69	20

GLOSSARY OF TERMS

Guide to Reading and Understanding Your Laboratory Report

The information below is designed to better explain the various terms used in your report of analytical results from the Laboratory. This is not intended as a comprehensive explanation, and if you have additional questions please contact your project representative.

Results Disclaimer - Information that may be provided by the customer, and contained within this report, include Permit Limits, Project Name, Sample ID, Sample Matrix, Sample Preservation, Field Blanks, Field Spikes, Field Duplicates, On-Site Data, Sampling Collection Dates/Times, and Sampling Location. Results relate to the accuracy of this information provided, and as the samples are received.

Abbreviations and Definitions

(dry)	Results are reported based on the dry weight of the sample. [this will only be present on a dry report basis for soils].
MDL	Method Detection Limit.
MDL (dry)	Method Detection Limit.
RDL	Reported Detection Limit.
RDL (dry)	Reported Detection Limit.
Rec.	Recovery.
RPD	Relative Percent Difference.
SDG	Sample Delivery Group.
U	Not detected at the Reporting Limit (or MDL where applicable).
Analyte	The name of the particular compound or analysis performed. Some Analyses and Methods will have multiple analytes reported.
Dilution	If the sample matrix contains an interfering material, the sample preparation volume or weight values differ from the standard, or if concentrations of analytes in the sample are higher than the highest limit of concentration that the laboratory can accurately report, the sample may be diluted for analysis. If a value different than 1 is used in this field, the result reported has already been corrected for this factor.
Limits	These are the target % recovery ranges or % difference value that the laboratory has historically determined as normal for the method and analyte being reported. Successful QC Sample analysis will target all analytes recovered or duplicated within these ranges.
Original Sample	The non-spiked sample in the prep batch used to determine the Relative Percent Difference (RPD) from a quality control sample. The Original Sample may not be included within the reported SDG.
Qualifier	This column provides a letter and/or number designation that corresponds to additional information concerning the result reported. If a Qualifier is present, a definition per Qualifier is provided within the Glossary and Definitions page and potentially a discussion of possible implications of the Qualifier in the Case Narrative if applicable.
Result	The actual analytical final result (corrected for any sample specific characteristics) reported for your sample. If there was no measurable result returned for a specific analyte, the result in this column may state "ND" (Not Detected) or "BDL" (Below Detectable Levels). The information in the results column should always be accompanied by either an MDL (Method Detection Limit) or RDL (Reporting Detection Limit) that defines the lowest value that the laboratory could detect or report for this analyte.
Uncertainty (Radiochemistry)	Confidence level of 2 sigma.
Case Narrative (Cn)	A brief discussion about the included sample results, including a discussion of any non-conformances to protocol observed either at sample receipt by the laboratory from the field or during the analytical process. If present, there will be a section in the Case Narrative to discuss the meaning of any data qualifiers used in the report.
Quality Control Summary (Qc)	This section of the report includes the results of the laboratory quality control analyses required by procedure or analytical methods to assist in evaluating the validity of the results reported for your samples. These analyses are not being performed on your samples typically, but on laboratory generated material.
Sample Chain of Custody (Sc)	This is the document created in the field when your samples were initially collected. This is used to verify the time and date of collection, the person collecting the samples, and the analyses that the laboratory is requested to perform. This chain of custody also documents all persons (excluding commercial shippers) that have had control or possession of the samples from the time of collection until delivery to the laboratory for analysis.
Sample Results (Sr)	This section of your report will provide the results of all testing performed on your samples. These results are provided by sample ID and are separated by the analyses performed on each sample. The header line of each analysis section for each sample will provide the name and method number for the analysis reported.
Sample Summary (Ss)	This section of the Analytical Report defines the specific analyses performed for each sample ID, including the dates and times of preparation and/or analysis.

Qualifier	Description
J	The identification of the analyte is acceptable; the reported value is an estimate.

ACCREDITATIONS & LOCATIONS

Pace Analytical National 12065 Lebanon Rd Mount Juliet, TN 37122

Alabama	40660	Nebraska	NE-OS-15-05
Alaska	17-026	Nevada	TN000032021-1
Arizona	AZ0612	New Hampshire	2975
Arkansas	88-0469	New Jersey–NELAP	TN002
California	2932	New Mexico [1]	TN00003
Colorado	TN00003	New York	11742
Connecticut	PH-0197	North Carolina	Env375
Florida	E87487	North Carolina [1]	DW21704
Georgia	NELAP	North Carolina [3]	41
Georgia [1]	923	North Dakota	R-140
Idaho	TN00003	Ohio–VAP	CL0069
Illinois	200008	Oklahoma	9915
Indiana	C-TN-01	Oregon	TN200002
Iowa	364	Pennsylvania	68-02979
Kansas	E-10277	Rhode Island	LA000356
Kentucky [1 6]	KY90010	South Carolina	84004002
Kentucky [2]	16	South Dakota	n/a
Louisiana	AI30792	Tennessee [1 4]	2006
Louisiana	LA018	Texas	T104704245-20-18
Maine	TN00003	Texas [5]	LAB0152
Maryland	324	Utah	TN000032021-11
Massachusetts	M-TN003	Vermont	VT2006
Michigan	9958	Virginia	110033
Minnesota	047-999-395	Washington	C847
Mississippi	TN00003	West Virginia	233
Missouri	340	Wisconsin	998093910
Montana	CERT0086	Wyoming	A2LA
A2LA – ISO 17025	1461.01	AIHA-LAP,LLC EMLAP	100789
A2LA – ISO 17025 [5]	1461.02	DOD	1461.01
Canada	1461.01	USDA	P330-15-00234
EPA–Crypto	TN00003		

[1] Drinking Water [2] Underground Storage Tanks [3] Aquatic Toxicity [4] Chemical/Microbiological [5] Mold [6] Wastewater n/a Accreditation not applicable

* Not all certifications held by the laboratory are applicable to the results reported in the attached report.

* Accreditation is only applicable to the test methods specified on each scope of accreditation held by Pace Analytical.

Internal Transfer Chain of Custody

Pace

☐ Rush Multiplier____X
☐ Samples Pre-Logged into eCOC

State Of Origin: KS
Cert. Needed: [X] Yes ☐ No

Workorder: 60465963 **Workorder Name:** COMBINED SEWER/J045885.01 **Owner Received Date:** 12/5/2024 **Results Requested By:** 12/19/2024

Report To	Subcontract To
Angie Brown Pace Analytical Kansas 9608 Loiret Blvd. Lenexa, KS 66219 Phone 1(913)563-1402	Pace National 12065 Lebanon Rd Mt. Juliet, TN 37122 Phone (615) 758-5858

Item	Sample ID	Sample Type	Collect Date/Time	Lab ID	Matrix	Preserved Containers: Unpreserved	Requested Analysis: Sulfide	LAB USE ONLY
1	S-2 10'-10.5'	PS	12/5/2024 09:00	60465963001	Solid	1	X	-01
2								
3								
4								
5								

L1010972

Transfers	Released By	Date/Time	Received By	Date/Time
1	JAP&G	12/6/24 1700		
2				
3			Christopher M. Kellum	12/7/24 0900

Comments

KS sample located 6091-R6-S2BS

PLEASE PULL TO SPLIT INTO JGFU

Cooler Temperature on Receipt _____ °C | **Custody Seal** Y or N | **Received on Ice** Y or N | **Samples Intact** Y or N

****In order to maintain client confidentiality, location/name of the sampling site, sampler's name and signature may not be provided on this COC document. This chain of custody is considered complete as is since this information is available in the owner laboratory.*

0.2+0.2

Sample Receipt Checklist

		If Applicable
COC Seal Present/Intact	Y N	
COC Signed/Accurate	Y N	VOA Zero Headspace Y N
Bottles arrive intact	Y N	Pres. Correct/Check Y N
Correct bottles used	Y N	
Sufficient volume sent	Y N	1 TOTAL
pH screen <0.5 mR/hr	Y N	

LR806972

Pace

Ship To:
Pace National
12065 Lebanon Rd
Mt. Juliet, TN 37122
Phone (615) 758-5858

INTER_LABORATORY WORK ORDER # 60465963
(To be completed by sending lab)

Sending Project No:	60465963
Receiving Project No:	
Check Box for Consolidated Invoice:	☐
Date Prepared:	12/06/24
REQUESTED COMPLETION DATE:	**12/19/2024**

Sending Region	IR60-Kansas	Sending Project Mgr.	Angie Brown
Receiving Region	IR850-Pace National	External Client	Universal Engineering_dba Geotechnology Inc
State of Sample Origin	KS	QC Deliverable	STD REPORT

All questions should be addressed to sending project manager.

Requested Reportable Units _____ Report Wet or Dry Weight? Wet Cert. Needed YES

WORK REQUESTED

Method Description	Container Type	Quantity of containers	Preservative	Quantity of Samples	Acode	Acode Desc
Sulfide	JGFU	1	Unpreserved	1	SI-21WET	SUB PASI WET

Special Requirements: Pace Regulatory Report (870)

FOR ANALYTICAL WORK COMPLETED THIS SECTION ALSO

Return Samples to Sending Region: ☐ Yes ☒ No

DISPOSITION of FORM

Original sent to the receiving lab - Copy kept at the sending lab

When work completed: Original sent to the ABM at the receiving laboratory. Copies are made to corporate as needed.

APPENDIX E

Pavement Core Photographs

B-1 **Combined Sewer Overflow Separation** J045885.01
Atchison, Kansas



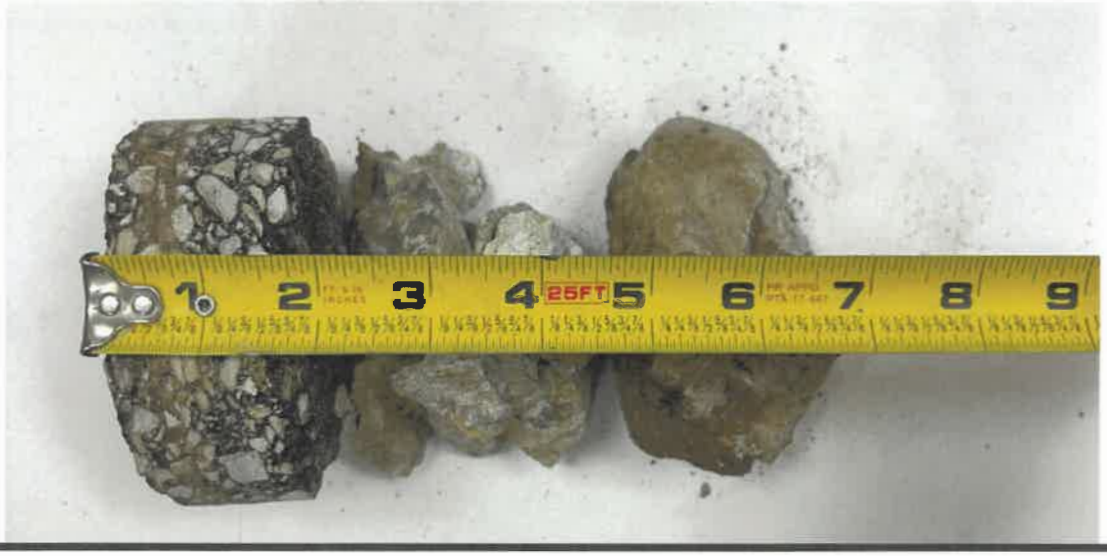
Asphalt	**Concrete**	**Granular Base**
2.5 inches	10 inches	None

B-2 **Combined Sewer Overflow Separation** J045885.01
Atchison, Kansas



Asphalt	**Brick**	**Concrete**	**Granular Base**
3.5 inches	2.5 inches	2 inches	None

B-3 **Combined Sewer Overflow Separation** J045885.01
Atchison, Kansas



Asphalt	**Concrete**	**Granular Base**
1.5 inches	5 inches (poor recovery)	None

APPENDIX F

Geotechnical Design Value Computation Sheets

MANHOLES

1) Net Allowable Bearing Pressure (q_a)

Net allowable bearing pressure is considered the additional pressure in excess of the weight of the existing soil.

- $q_a = \frac{q_{ultimate}}{3}$ where 3 is the factor of safety

- Based on the results of the boring, assume that the manhole is bearing on stiff fine-grained soil or medium dense sand.

- For stiff fine-grained soil,

$$q_{ult} = c N_c s_c + \bar{q} N_q + 0.5 \gamma B N_\gamma s_\gamma$$

Per Terzaghi assuming $Q' = \text{zero}$
$N_c = 5.7 \quad N_q \doteq 1.0 \quad N_\gamma = \text{zero}$

Assume $c = 1{,}000$ psf (see log of B-2 and Sample ST4)
$\bar{q} = \gamma \times \Delta$ where Δ = frost depth = 3 feet
$= 120$ pcf $\times$ 3 feet $= 360$ psf
$s_c = 1.0$ (for strip footings)

MANHOLES (cont.)

q_{ult} for fine-grained soil $= (1{,}000)(5.7)(1.0) + 360(1.0) + \text{zero}$

$= 5{,}700 + 360$

$= 6{,}060$ psf

Net Ult. Bearing Capacity for Fine-Grained Soil $= q_{ult} - \gamma \times D$

$= 6{,}060 - 360$

$= 5{,}700$ psf

Net Allowable Bearing Capacity for Fine-Grained Soil $= \frac{5{,}700}{3}$ where 3 is the factor of safety

$= 1{,}900$ psf for strip footing

Net Allowable Bearing Capacity for Fine-Grained Soil for MANHOLES $= 1{,}900 \times 1.3$

$\doteq 2{,}500$ psf

Note: a design value of 1,500 psf is recommended to limit manhole settlement to 1/4" or less and is based on the author's experience.

MANHOLES (cont.)

- For medium dense sand

$$q_{ult} = cN_c s_c + \bar{q} N_q + 0.5 \gamma B N_\gamma s_\gamma$$

since the bearing material is sand, c = zero

Assume sand w/ an effective friction angle of 28°

Per Terzaghi, sand w/ $\phi' = 28° \rightarrow N_q = 14.21$

$N_c = 27.09$

$N_\gamma = 11.7$

$\bar{q} = \gamma \times \Delta$ where Δ = frost depth = 36 inches

$= 120 pcf \times 3 feet = 360 psf$

B = typical manhole width = 42 inches

$S_\gamma = 1.0$ for strip footings

$$q_{ult(sand)} = cN_c s_c + \bar{q} N_q + 0.5 \gamma B N_\gamma s_\gamma$$

$$= 0 + 360(14.21) + 0.5(120 pcf)(3.5')(11.7)(1.$$

$$= 0 + 5,116 + 2,457$$

$$= 7,573 psf$$

Net Ult. Bearing Capacity = $q_{ult\,sand} - \gamma \times \Delta$

$= 7,573 psf - 360 psf$

$\approx 7,200 psf$

MANHOLES (cont.)

$$\text{Net Allowable Bearing Capacity of Sand} = \frac{q_{ult.sand(net)}}{3}$$ where 3 is factor of safety

$$= \frac{7,200}{3}$$

$$= 2,400 \text{ psf}$$

Note: A design value of 1,500 psf is recommend to limit manhole settlement to 1/4" or less and is based on the author's experience.

MANHOLES (cont.)

- Uplift resistance $= 0.7 \times \frac{C}{\text{Factor of Safety}}$

$= 0.7 \times \frac{1{,}000 \text{ psf}}{2}$

Allowable Uplift Resistance $= 350$ psf

Lateral Earth Pressures

1) Fine-grained soil specified in the report;
Assume $\phi' = 27°$ and that the soil is compacted to 120pcf

At-rest Lateral Earth Pressure $= K_o \times \gamma_{soil}$
$= (1-\sin\phi') \times \gamma_{soil}$
$= (1-\sin 27°) \times 120\,pcf$
$= 0.55 \times 120\,pcf$
$\cong 65\,pcf$

Active Lateral Earth Pressure $= K_a \times \gamma_{soil}$
$= \left(\frac{1-\sin\phi'}{1+\sin\phi'}\right) \times \gamma_{soil}$
$= \left(\frac{1-\sin 27°}{1+\sin 27°}\right) \times 120\,pcf$
$= 0.38 \times 120\,pcf$
$\cong 45\,pcf$

Lateral Earth Pressures (continued)

2) Well-graded Crushed Rock; Assume $\phi' = 37°$ and that the crushed rock is compacted to 140 pcf

At-Rest Lateral Earth Pressure $= K_o \times \gamma_{rock}$

$= (1 - \sin\phi') \times \gamma_{rock}$

$= (1 - \sin 37°) \times 140\ pcf$

$= 0.40 \times 140\ pcf$

$\doteq 56\ pcf$

Active Lateral Earth Pressure $= K_a \times \gamma_{rock}$

$= \left(\frac{1-\sin\phi'}{1+\sin\phi'}\right) \times \gamma_{rock}$

$= \left(\frac{1-\sin 37°}{1+\sin 37°}\right) \times 140\ pcf$

$= 0.25 \times 140\ pcf$

$\doteq 35\ pcf$

Table A.1 Soil-test evaluation

Soil Characteristics Based on Samples Taken Down to Pipe Depth	Points*
Resistivity—ohm-cm (based on water-saturated soil box)	
<1,500	10
≥1,500–1,800	8
>1,800–2,100	5
>2,100–2,500	2
>2,500–3,000	1
>3,000	0
pH:	
0–2	5
2–4	3
4–6.5	0
6.5–7.5	0†
7.5–8.5	0
>8.5	3
Redox potential:	
> +100 mV	0
+50 to +100 mV	3.5
0 to +50 mV	4
Negative	5
Sulfides:	
Positive	3.5
Trace	2
Negative	0
Moisture:	
Poor drainage, continuously wet	2
Fair drainage, generally moist	1
Good drainage, generally dry	0

*Ten points indicates that soil is corrosive to ductile-iron pipe; protection is needed.

†If sulfides are present and low or negative redox-potential results are obtained, add three points for this range.

B-2, 10 feet
225
5.8
231.9
Undetected
Total = 11

8. *Oxidation-reduction (redox) potential.* The redox potential of a soil is significant, because the most common sulfate-reducing bacteria can live only in anaerobic conditions. A redox potential greater than +100 mV demonstrates that the soil is sufficiently aerated, preventing sulfate reducers from forming. Potentials of 0 to +100 mV may or may not indicate anaerobic conditions; however, a negative redox potential definitely indicates the anaerobic conditions in which sulfate reducers thrive. The redox test is performed using a pH/mV meter with a combination ORP electrode inserted into the soil sample. It should be noted that soil samples removed from a boring or excavation can undergo a change in redox potential when exposed to

Pg 8